

F-doping induced interfacial built-in electric field and charge redistribution construction in CoS@NiCoLDH p-p heterojunctions for high-performance supercapacitors --Manuscript Draft--

Manuscript Number:	CEJ-D-26-17720
Article Type:	Research Paper
Section/Category:	Novel Materials for Energy and Advanced Applications
Keywords:	F-doping; Built-in electric field; p-p heterojunction; Supercapacitors; Boost carrier density
Abstract:	The synergistic regulation of interface engineering and defect engineering is pivotal for overcoming the sluggish ion diffusion kinetics in supercapacitors and realizing high-performance electrode materials. A novel strategy is proposed to construct a nickel-cobalt layered double hydroxide (NiCo-LDH) with rapid reaction kinetics by simultaneously introducing F-doping and establishing a p-p heterojunction. The synthesized electrode material features a hierarchical nanoflower like architecture, where the interlaced nanosheets facilitate the construction of active interfaces and the establishment of built-in electric field (IEF). Experimental characterization and DFT modeling further demonstrate that F-doping modulates the electronic environment to effectively boost carrier density. Meanwhile, the IEF at the p-p interface, induced by the work function difference (4.57 eV $\rightarrow$ 5.23 eV), provides a potent driving force for the directional migration of holes, thereby optimizing the charge transfer energy barrier. Benefiting from such synergistic electronic regulation, the prepared electrode material displays an impressive specific capacitance value of 3442 F g⁻¹ (1 A g⁻¹) and maintains exceptional rate capability. This work yields a profound understanding of the intrinsic mechanism of the synergistic interplay between F-doping and heterojunction interfaces, offering a valuable strategy for developing superior energy storage materials.

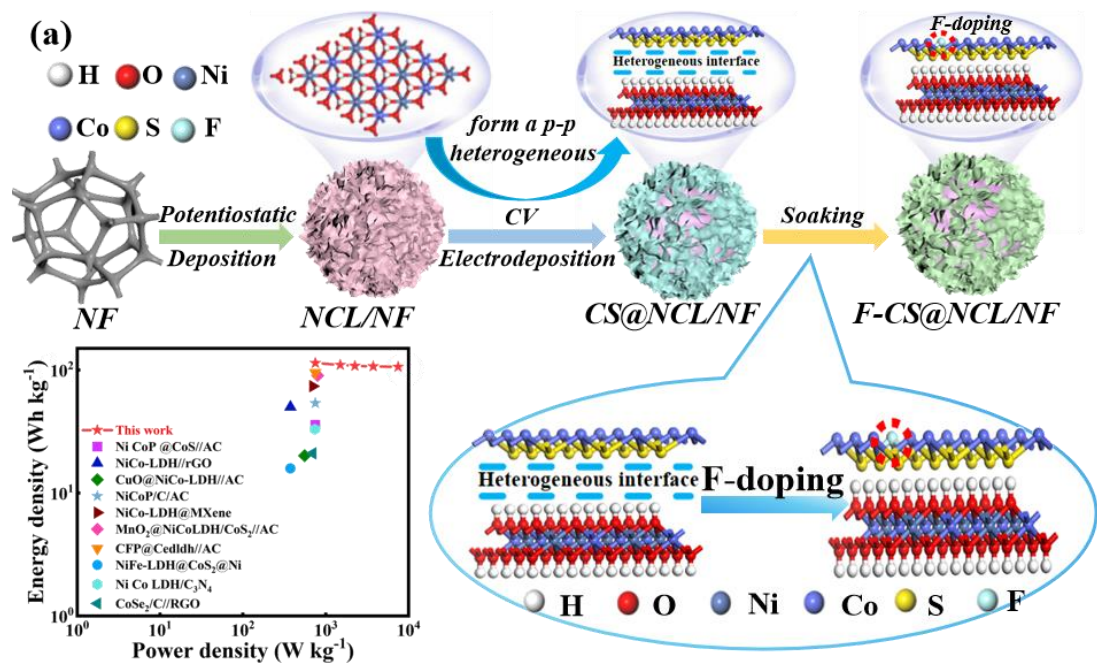
F-doping induced interfacial built-in electric field and charge redistribution construction in CoS@NiCoLDH p-p heterojunctions for high-performance supercapacitors

Zihan Zhang, Qianqian Shan, Kairan Hu, Jie Yang, Jibo Jiang*, Sheng Han*

Department of Chemical Engineering and Energy Technology, Shanghai Institute of Technology, Shanghai 201418, China.

*Corresponding authors.

E-mail addresses: jibojiang0506@163.com (J. Jiang), hansheng654321@sina.com (S. Han).



F-doping induced interfacial built-in electric field and charge redistribution construction in CoS@NiCoLDH p-p heterojunctions for high-performance supercapacitors

Zihan Zhang, Qianqian Shan, Kairan Hu, Jie Yang, Jibo Jiang*, Sheng Han*

Department of Chemical Engineering and Energy Technology, Shanghai Institute of Technology, Shanghai 201418, China.

*Corresponding authors.

E-mail addresses: jibojiang0506@163.com (J. Jiang), hansheng654321@sina.com (S. Han).

Highlights

- Construction of F-doped CoS@NiCoLDH p-p heterojunction optimizes the interfacial electronic structure.
- Interfacial band bending induces a strong built-in electric field (IEF) for directional carrier migration.
- Theoretical calculations reveal F-induced charge redistribution lowers the charge transfer barrier.
- The p-p interface synergistically modulates the d-band center and enhances the reaction kinetics.

F-doping induced interfacial built-in electric field and charge redistribution construction in CoS@NiCoLDH p-p heterojunctions for high-performance supercapacitors

Zihan Zhang, Qianqian Shan, Kairan Hu, Jie Yang, Jibo Jiang*, Sheng Han*

Department of Chemical Engineering and Energy Technology, Shanghai Institute of Technology, Shanghai 201418, China.

*Corresponding authors.

E-mail addresses: jibojiang0506@163.com (J. Jiang), hansheng654321@sina.com (S. Han).

Abstract: The synergistic regulation of interface engineering and defect engineering is pivotal for overcoming the sluggish ion diffusion kinetics in supercapacitors and realizing high-performance electrode materials. A novel strategy is proposed to construct a nickel-cobalt layered double hydroxide (NiCo-LDH) with rapid reaction kinetics by simultaneously introducing F-doping and establishing a p-p heterojunction. The synthesized electrode material features a hierarchical nanoflower like architecture, where the interlaced nanosheets facilitate the construction of active interfaces and the establishment of built-in electric field (IEF). Experimental characterization and DFT modeling further demonstrate that F-doping modulates the electronic environment to effectively boost carrier density. Meanwhile, the IEF at the p-p interface, induced by the work function difference ($4.57\text{ eV} \rightarrow 5.23\text{ eV}$), provides a potent driving force for the directional migration of holes, thereby optimizing the charge transfer energy barrier. Benefiting from such synergistic electronic regulation, the prepared electrode material displays an impressive specific capacitance value of 3442 F g^{-1} (1 A g^{-1}) and maintains exceptional rate capability. This work yields a profound understanding of the intrinsic mechanism of the synergistic interplay between F-doping and heterojunction interfaces, offering a valuable strategy for developing superior energy storage materials.

Keywords: F-doping, Built-in electric field, p-p heterojunction, Supercapacitors, Boost carrier density

1. Introduction

Driven by energy crises and environmental concerns, developing efficient energy storage systems is crucial for scaling renewable energy [1]. Supercapacitors, distinguished by superior power density, rapid charge-discharge rates, superior cyclability, as well as exemplary safety, have emerged as favorable candidates in the advanced energy storage landscape [2]. Nevertheless, their widespread

application remains severely constrained by an intrinsic deficiency in energy density [3]. Since active materials dictate supercapacitor performance, current research prioritizes designing novel electrodes with high capacity, excellent conductivity, and robust stability. Aligned with this design philosophy, transition metal compounds endowed with abundant redox-active sites are viewed as ideal building blocks for next-generation high-performance supercapacitors, owing to their ability to deliver substantial Faradaic pseudocapacitance [4].

Thanks to their structural advantages, LDHs stand out among transition metal compounds as ideal candidates for high-performance energy storage. Benefiting from their distinctive 2D layered structure and large interlayer spacing, these materials provide stable support that facilitates rapid ion transport and exchange during redox reactions [5]. Within the diverse LDH family, bimetallic NiCo-LDH distinguishes itself from monometallic counterparts or combinations containing inactive components (e.g., Zn/Ni-based LDHs) by demonstrating ultrahigh theoretical specific capacity and rich redox chemistry [6]. This superiority stems primarily from the synergistic interaction between Ni and Co species, where overlapping potential windows facilitate continuous multistep redox reactions, thereby effectively multiplying active sites and imparting exceptional pseudocapacitive properties. Despite these merits, LDHs frequently suffer from restricted electron transport under high-rate or long-cycling conditions, stemming from intrinsic sluggish charge transfer kinetics that hampers the directional migration of carriers [7].

Coupling LDHs with transition metal sulfides (TMSs) to construct heterostructures represents a potent strategy to surmount these electronic transport limitations, leveraging the high conductivity and complementary electronic structures of TMSs. Among various TMSs, cobalt sulfide (CoS) garners significant attention in the supercapacitor domain for its metallic-like conductivity, superior chemical stability, and prolific redox-active sites [8]. Although morphological engineering has yielded some improvements in its electrochemical behavior [9,10], single-component CoS remains constrained by limited interfacial charge modulation capabilities and reaction kinetics. Crucially, the built-in electric field (IEF) established within the interfacial regions of heterojunctions electrodes serves as a pivotal physical mechanism for governing carrier behavior. By driving interfacial charge redistribution and accelerating charge transfer, the IEF provides a theoretical cornerstone for high-performance interface design. Upon contact, differences in Fermi levels or energy band structures between heterogeneous components spontaneously induce charge redistribution, culminating in a

stable interfacial IEF [2,11]. Consequently, in situ coupling of CoS with Ni Co-LDH to form a heterojunction system not only enhances the electron transport capacity of LDHs but also synergistically accelerates reaction kinetics.

Beyond structural interface engineering, heteroatom doping offers a precise avenue for modulating the electronic structure at the heterointerface. Among potential dopants, the fluorine (F) atom, characterized by its extreme electronegativity, can significantly alter local potential distributions and energy level alignments without compromising host lattice stability [12]. This augmented interfacial driving force facilitates directional charge migration and lowers the energy barrier for charge transport, thereby intrinsically accelerating reaction kinetics [13]. Ultimately, the synergistic interplay between F-doping and heterostructure engineering optimizes the interfacial electronic structure while sustaining stable electrochemical behavior sustaining stable electrochemical behavior, realizing synchronous enhancements in both energy and power densities.

Guided by these principles, the current work presents an F-doped surface-decorated F-CoS @NiCoLDH/NF heterojunction electrode fabricated on a nickel foam (NF) scaffold by employing a sequential electrodeposition route alongside chemical immersion strategy. By synergistically introducing band matching effects and IEF driving forces at the heterointerface, the resulting architecture establishes continuous, high-efficiency charge transport channels within the electrode, significantly optimizing reaction kinetics and ion diffusion. Moreover, density functional theory (DFT) simulations clarify that F-doping effectively tunes and intensifies the IEF effect at the heterointerface by regulating the energy level alignment and Fermi level positions on both sides, which synergistically lowers the interfacial charge transfer barrier. Driven by the interfacial electronic synergy, the F-CS@NCL/NF//AC HSC delivers 114 W h kg^{-1} at 750 W kg^{-1} , demonstrating its huge potential for advanced energy storage.

2. Experimental section

2.1. Materials

Analytical-grade $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, thiourea and NH_4F were procured from Shanghai Titan Technology Co., Ltd. and used as received.

2.2. Materials Synthesis

2.2.1. Preparation of NiCoLDH/NF (NCL/NF)

4 mmol of $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ were dissolved in 100 mL DIW and stirred to dissolve. NF acted as the working electrode, coupled with a Pt wire counter electrode and an Hg/HgO reference electrode. A potentiostatic method was executed applying a fixed potential of -2 V lasting 20 min. The resulting samples were washed thoroughly before being dried under vacuum for 12 h.

2.2.2. Synthesis of CoS@NiCoLDH/NF (CS@NCL/NF)

2 mmol of $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and 2 mmol of thiourea were dissolved in 100 mL DIW and stirred to dissolve. CV electrodeposition was then performed on the NCL/NF electrode (-1.2 to -0.2 V). After washing thrice with DIW, the sample was dried under vacuum at 60°C for 12 h to yield CS@NCL/NF . For the synthesis of the pure CoS sample, bare NF was used as the working electrode while maintaining other parameters constant.

2.2.3. Synthesis of F-CoS@NiCoLDH/NF (F-CS@NCL/NF)

The as-prepared CS@NCL/NF was immersed into purified water containing 2 mmol of NH_4F . The F-CS@NCL/NF product was yielded after 3 hours of immersion. The F-CoS samples were prepared by substituting the precursor with CoS and repeating the aforementioned immersion procedure.

2.2.4. Synthesis of the F-CoS@NiCoLDH/NF Hybrid Supercapacitor (F-CS@NCL//AC HSC)

A mixture of AC, acetylene black, and PTFE (8:1:1 weight ratio) was thoroughly ground and coated onto NF. Vacuum dry at 60°C for 6 h. By sandwiching the F-CS@NCL/NF cathode along with the activated carbon anode on opposite sides of a cellulose separator followed by compression encapsulation.

2.3. Material Characterization

To resolve the intricate nano structural characteristics, the microscopic morphology and fine nanostructures of the products were observed on a ZEISS Gemini SEM as well as a JEOL JEM-2100F TEM. Concurrently, elemental mapping analysis was executed via the attached EDS apparatus to verify the coexistence and dispersion of constituent elements in the prepared samples. X-ray diffraction (XRD) was employed to analyze the crystal structure and phase composition of the samples, using an X'Pert PRO MPD instrument. device. Subsequently, XPS analysis was carried out to identify the constituents, oxidation states, and binding properties of the samples. Furthermore, the types of functional groups contained within the specimens were analyzed in detail across a

wavenumber range of 500 to 3900 cm^{-1} with a BRUKER TENSOR 37 Fourier–transform infrared spectrometer (FTIR).

2.4. Electrochemical Characterization

Electrochemical assessments, encompassing CV, galvanostatic charge–discharge (GCD), and electrochemical impedance spectroscopy (EIS), were executed in a three–electrode configuration utilizing the 1 cm^2 in situ grown sample on NF as the working electrode. The setup included an Hg/HgO reference electrode and a Pt counter electrode immersed in an alkaline electrolytic medium. To evaluate the electrochemical behavior, CV profiles were collected between 0 and 0.5 V with sweep rates varying from 5 to 50 mV s^{-1} . The potential window for CV testing was spanning 0–0.5 V at sweep rates extending from 5 to 50 mV s^{-1} . GCD profiles were collected at 1–20 A g^{-1} within a 0–0.4 V range, while EIS data were collected in the 10^{-2} – 10^{-5} Hz frequency range with a 5 mV AC perturbation under open–circuit conditions. The electrochemical storage metrics, including capacitance (F g^{-1}) and specific capacity (C g^{-1}), were then derived from the GCD plots via Equations 1 and 2.

$$C = \frac{I \Delta t}{m \Delta V} \quad (1)$$

$$Q = \frac{I \Delta t}{3.6m} \quad (2)$$

Specifically, I and Δt denote the applied current (A) and discharge period (s), while ΔV and m represent the mass of the voltage range (V) and active species (g). To ensure electrochemical balance, the mass loading ratio of the positive to negative electrodes was optimized following the relationship shown in Equation 3.

$$\frac{m_+}{m_-} = \frac{C_- \Delta V_-}{C_+ \Delta V_+} \quad (3)$$

The charge balance relationship stipulates that C^+ and C^- correspond to the specific capacitances of the positive and negative electrodes respectively while ΔV^+ and ΔV^- represent their corresponding discharge voltage ranges. Accordingly, the AC electrode loading was set to 5–6 mg cm^{-2} . The electrochemical performance metrics, specifically energy density (E , W h kg^{-1}) and power density (P , W kg^{-1}), were then quantified by Equations 4 and 5.

$$P = \frac{3600 E}{\Delta t} \quad (4)$$

$$E = \frac{C \Delta V^2}{7.2} \quad (5)$$

where C represents the specific capacitance value, while Δt and ΔV indicate period of discharge during the measurement and the potential drop.

2.5. DFT Calculation

First-principles investigations relying on DFT were executed using the Materials Studio package software suite which was primarily employed for the construction of molecular models and the precise optimization of crystalline structures. Throughout the geometric optimization, the convergence criteria were strictly defined by interatomic force fluctuations below 0.03 eV/Å alongside an energy variation threshold lower than 10^{-5} eV. The optimization process was only considered converged when both requirements were simultaneously satisfied by the calculation results. For the crystalline structures of NCL and CoS, the corresponding supercell models were constructed via slicing treatments, and accurate lattice constants were subsequently obtained upon the completion of the optimization. First-principles simulations were conducted using the VASP code, adopting the projector augmented wave (PAW) method and a plane-wave basis with a 450 eV energy cutoff. Based on this theoretical framework, we systematically investigated the differential charge density, density of states (DOS), and band structures of the materials. These investigations aimed to provide a fundamental and in-depth explanation regarding the electronic properties as well as the internal charge distribution patterns of the materials.

3 Results and Discussion

3.1. Morphology and structure characterizations

The formation pathway for the F-CS@NCL/NF heterojunction as followed Fig. 1a. Initially, the NCL/NF nanostructure was grown on a NF substrate through potentiostatic deposition utilizing $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$. This structure served as a scaffold for the deposition of CoS via CV to establish the CS@NCL/NF heterointerface. Ultimately, heteroatom doping was accomplished through the introduction of ammonium fluoride to yield the targeted F-CS@NCL/NF product. In a comparative experiment, pure CoS/NF was synthesized by substituting the working electrode with bare NF. Notably, the morphology remains largely unaltered after F-doping, signifying that the doping strategy optimizes the electronic configuration without compromising the inherent crystal growth mechanism.

SEM images reveal that NCL/NF features ultra-thin cluster microspheres assembled from nanosheets (Fig. 1b and Fig. 1c), providing abundant anchors for active species. Upon CoS

introduction, the morphology evolves into interlaced vertical nanosheet arrays (Fig. 1d and Fig. 1e). Compared with pure CoS/NF (Fig. S1) also presents similar sheet-like characteristics, confirming that the morphology is dominated by the growth habit of CoS/NF [14]. Specifically, the NCL/NF clusters serve as a skeleton to induce the uniform nucleation of CoS/NF at the initial stage of deposition. subsequently, CoS/NF extends outward to form well-defined and extended sheet-like structures covering the NCL/NF surface. This hierarchical architecture provides open ion transport channels and ensures intimate interfacial contact, facilitating efficient charge transfer [4]. The well-preserved morphology after F-incorporation (Fig. 1d, Fig. 1e, Fig. 1f and Fig. 1g) implies that the doping process exerts a negligible impact on the intrinsic crystallization pathways, ensuring consistent nucleation and growth kinetics [2]. The microscopic architecture of the F-CS@NCL/NF composite was elucidated utilizing TEM analysis. In Fig. 1h, NCL is visualized alongside CoS, well-integrated heterostructure. Lattice fringes with d-spacings of 0.294 nm and 0.386 nm are observed, corresponding to the (100) and (006) planes of CoS and NCL, respectively (Fig. 1i). The results corroborate the XRD findings and support the existence of a CoS/NCL heterointerface. Furthermore, EDS mapping highlights the even dispersion of Co, Ni, F, S, C, and O elements across the sample, confirming the successful preparation of F-CS@NCL/NF.

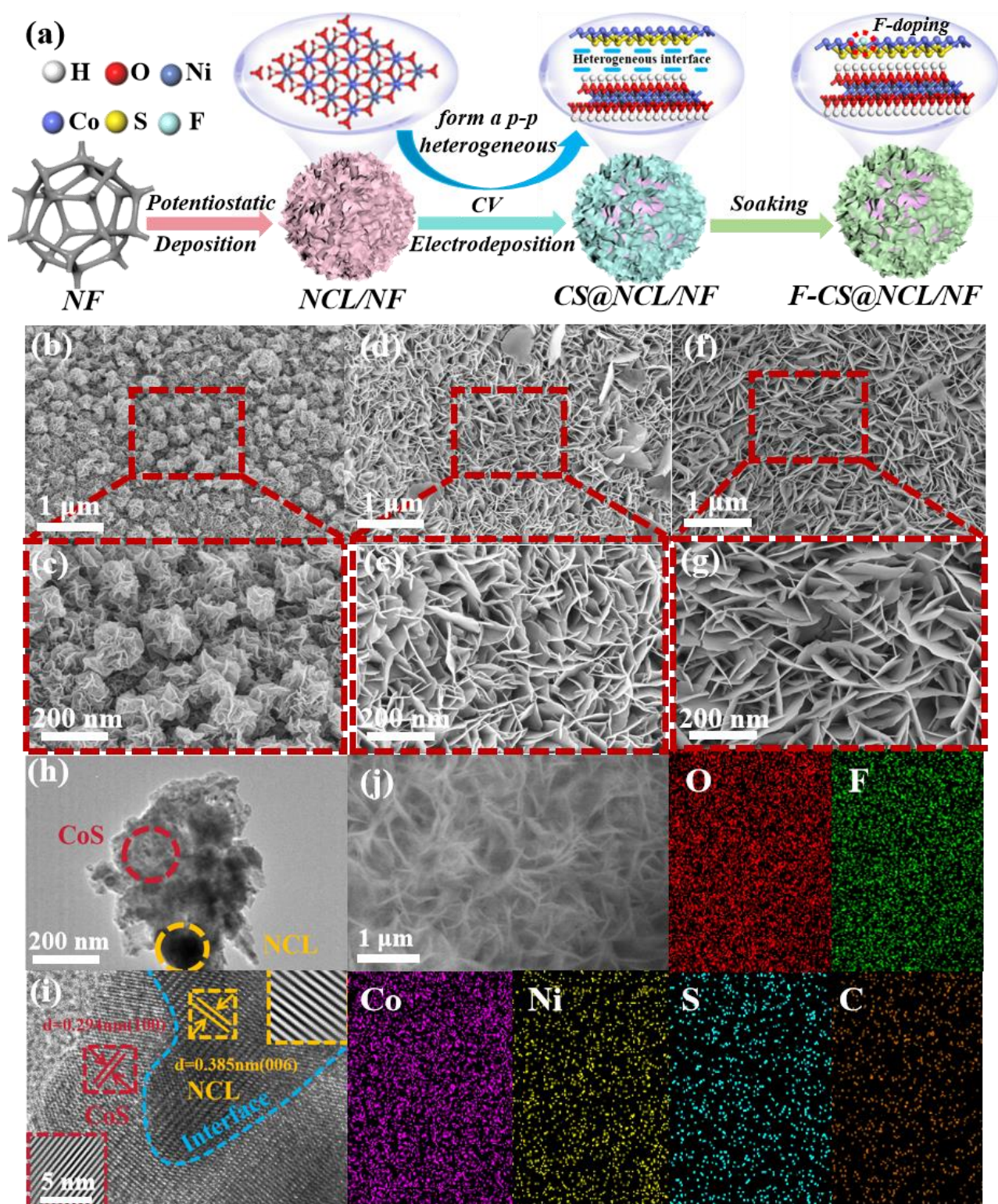


Fig. 1. (a) Schematic diagram depicting the preparation route for F-CS@NCL/NF. SEM micrographs corresponding to (b–c) NCL, (d–e) CS@NCL/NF, and (f–g) F-CS@NCL/NF. (h) TEM images of F-CS@NCL/NF. (i) HRTEM images of F-CS@NCL/NF. (j) HAADF image and corresponding element mapping images of F-CS@NCL/NF.

After in-depth understanding of the material microstructure, electrochemical Mott–Schottky

measurements were conducted to further determine the electrical properties, which notably verified the successful construction of the p-p heterojunction. Figs. 2a, Fig. 2b, Fig. 2c and Fig. 2d illustrate the structural models of NCL, CoS, CS@NCL and F-CS@NCL employed for DFT calculations. The semiconducting nature of these materials is substantiated by the energy band structure analysis (Figs. 2e, Fig. 2f, Fig. 2g and Fig. 2h), where the bandgap serves as a metric to evaluate conductivity with the Fermi level marked by red dashed lines. The results indicate that all samples exhibit semiconducting properties, which aligns well with the results obtained from M-S measurements [15]. The CS@NCL (Fig. 2f) composite exhibits a denser band structure and narrower bandgap than individual NCL (Fig. 2e) and CoS (Fig. 2h), favoring electron excitation. Remarkably, the bandgap of F-CS@NCL (Fig. 2g) further contracts relative to CS@NCL, with the energy bands exhibiting a shifting trend toward the Fermi level as illustrated in Fig. 2g. Concurrently, the bandgap of F-CoS (Fig. 2j) is also narrower than that of pristine CoS. These observations suggest that F-doping leads to a marked improvement in the material's conductivity, which is attributed to a diminished energy barrier for electron transition. Furthermore, F-doping induces a collective shift of the conduction and valence bands toward the Fermi level while substantially enhancing the DOS. Such electronic structure tuning effectively facilitates the dynamic attachment and detachment of ions from the electrolyte onto the electrode interface.

We characterized the electrical properties through Mott-Schottky (M-S) measurements, aiming to demonstrate the existence of the p-p type heterostructure. Figs. S2d and Fig. S2e reveals that the M-S plots of both NCL/NF and CoS/NF exhibit distinct negative slopes that align with the typical behavior of p-type semiconductors, thereby confirming the establishment of a p-p heterojunction [16]. Furthermore, as indicated in Figs S2f-S2h, F-CoS/NF, CS@NCL/NF, and F-CS@NCL/NF also display negative slopes, adhering to the fundamental physics of semiconductor contacts. M-S plots (Fig. 2i) confirm F-CS@NCL/NF exhibits the lowest slope absolute value, indicating maximum carrier density. The resulting transition effectively mitigates the space charge transport resistance and significantly enhances the hole migration rate, consequently improving the electrochemical charge storage properties of the material. Verification of the computational outcomes was performed via CV in conjunction with constant current charge/discharge (GCD) analyses. Meanwhile, nearly the flat-band potential difference ($|\Delta E_{fb}|$) of all samples was observed within the 0.2–0.4 V range (Figs S2d-S2h) indicates the formation of an optimal interfacial potential gradient.

The established potential gradient not only provides sufficient driving force for the directional transport of holes but also effectively suppresses carrier recombination [17].

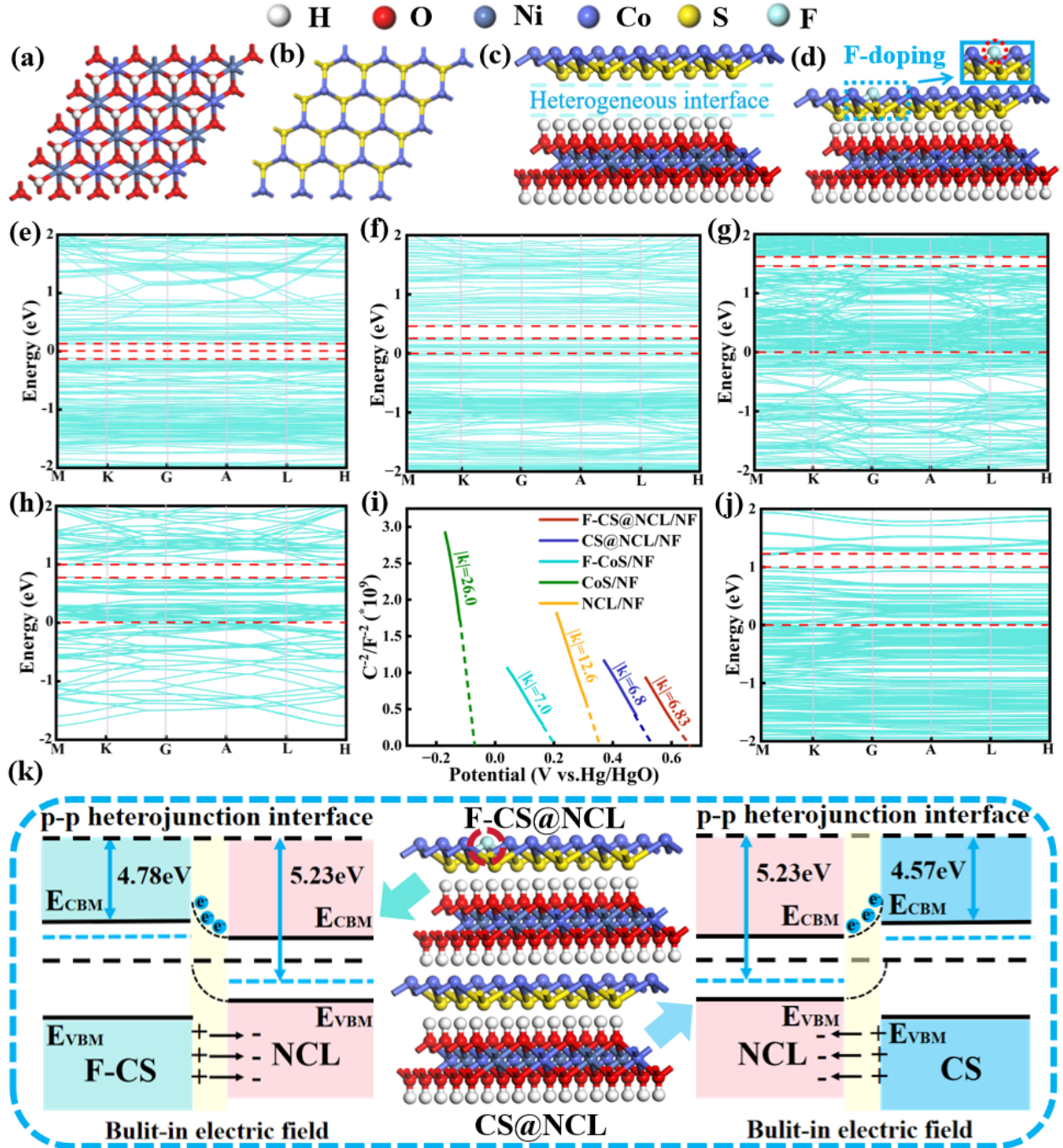


Fig. 2. Structural models of (a) NCL, (b) CoS, (c) CS@NCL, and (d) F-CS@NCL. Band structure plots of (e) NCL, (f) CS@NCL, (g) F-CS@NCL, (h) CoS, (j) F-CoS. (i) Absolute Value of Slope Comparison Diagram for Mott-Schottky curves of NCL/NF, CoS/NF, F-CoS/NF, CS@NCL/NF, F-CS@NCL/NF. (k) Illustration depicting the electron transfer mechanism.

We investigated the charge transfer mechanism at the hetero-interface by evaluating the work functions (Φ) of NCL, CoS, and F-CoS (Figs. S2f, Fig. S2g and Fig S2h). Representing the energy

difference between the Fermi level and the vacuum level, the work functions for the three materials were determined to be 5.23, 4.57, and 4.78 eV, respectively. Due to the distinct work function differences between phases, the spontaneous alignment of Fermi levels triggers band bending at the interface, resulting in the generation of a space charge region and an interfacial potential barrier, which lays the thermodynamic foundation for the effective establishment of the p–p heterojunction. As evidenced by the charge distribution model presented in Fig. 2k, NCL serves as an electron acceptor because of its higher work function, a characteristic that aligns with the classic high–work–function core and low–work–function shell configuration for p–p heterojunctions [18]. Furthermore, the work function gradient between F–CoS and CoS further drives electron transfer toward NCL. This potential difference acts as the core factor that induces interfacial charge redistribution. The work function gradient ($\text{NCL} > \text{F–CoS} > \text{CoS}$) induces interfacial charge redistribution, consistent with the M–S flat–band potential trends.

XRD patterns (Fig. 3a) reveal the diffraction peaks appearing at 12.35° , 25.35° , 33.46° , and 59.26° are indexed to the (003), (006), (012), and (110) crystallographic planes of NCL (PDF#38–0715) [19]. The diffraction peaks at 30.63° , 35.37° , 36.60° , 46.93° , 54.44° , and 62.15° correspond to the (100), (002), (101), (102), (110), and (103) crystallographic planes of CoS (PDF#65–3418), respectively [20]. The XRD pattern of F–CS@NCL/NF maintains the host framework of CS@NCL/NF with slightly attenuated intensity, indicating preserved crystallinity despite F–doping.

FTIR reveal the characteristic bands at $3265\text{--}3621$ and 1632 cm^{-1} attribute the vibrations of surface water or hydroxyl groups (Fig. 3b) [21]. In the spectra containing NCL/NF, the peak at 1384 cm^{-1} substantiate the typical vibration of interlayer NO_3^- [22]. Specifically, the peaks at 1350 , 1150 and 1032 cm^{-1} correspond to the M–OH vibrational modes [23]. The fingerprint region analysis indicates that the feature at 788 cm^{-1} is assigned to the Co–S bond [24]. Following the construction of the heterojunction, the significant shifts of the M–O and Co–S bonds verify the profound charge redistribution at the interface rather than a mere physical stacking [25]. Upon the further introduction of F–doping, the characteristic peak intensities of F–CS@NCL/NF exhibit a reasonable attenuation and the Co–S bond displays a distinct blue shift which reveals the effective regulation of the interfacial energy level structure by the highly electronegative F atoms [26].

Raman spectra (Fig. 3c) display characteristic Co–O (556 cm^{-1}) and Ni–O (512 cm^{-1}) modes in NCL/NF, and the A1g mode of CoS/NF (358 cm^{-1}) [5]. The spectrum of CS@NCL/NF concurrently

preserves the characteristic peaks associated with each constituent phase, thereby validating the successful construction of the p-p heterojunction. F-CS@NCL/NF exhibits a discernible redshift and reduced intensity, Such a transformation stems from the pronounced electronegativity of F, which triggers lattice distortion and atomic rearrangement [27]. Furthermore, the intensity attenuation is associated with doping-induced lattice defects and increased carrier concentration, which is consistent with the lattice expansion and enhanced conductivity revealed by XRD and M-S measurements. Ultimately, the synergistic optimization of vibrational modes and electronic configurations facilitates accelerated charge transfer kinetics during electrochemical reactions [14].

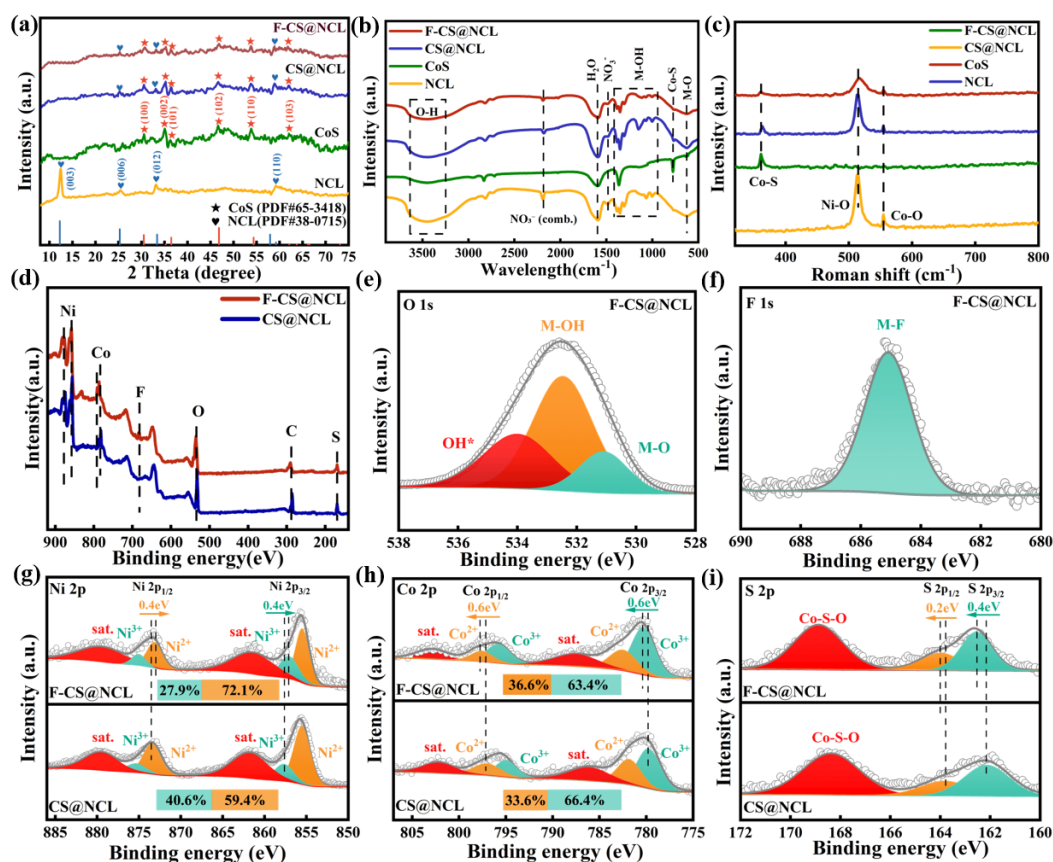


Fig. 3. (a) XRD profiles, (b) FTIR curves, and (c) Raman characterization of NCL/NF, CoS/NF, CS@NCL/NF, and F-CS@NCL/NF. (d) XPS of CS@NCL/NF and F-CS@NCL/NF. High-resolution XPS core-level spectra for (e) O 1s, (f) F 1s, (g) Ni 2p, (h) Co 2p, and (i) S 2p obtained from CS@NCL/NF and F-CS@NCL/NF.

Surface chemical environments and elemental constituents of F-CS@NCL/NF were probed using XPS. The presence of Ni, Co, F, O, C, and S is clearly evidenced by their specific signals in the survey spectrum (Fig. 3d), confirming the elemental composition of the electrode material. Subsequently, the F 1s spectrum presented in Fig. 3f reveals a characteristic peak at 685.3 eV

assigned to the metal–F bond, verifying that the F element has been incorporated into the material system through chemical bonding [28]. From the O 1s high-resolution spectrum (Fig. 3e), the three characteristic peaks after deconvolution are located at 533.98, 532.48, and 531.08 eV, respectively. These features are assigned to surface-adsorbed hydroxyl groups (–OH), M–OH bonds in NCL, and lattice oxygen (M–O) in sequence, confirming the synthesis of the target metal hydroxides [29]. To uncover the role of F-doping and the construction of the p–p heterojunction on the electronic states of the metal centers, we conducted an in-depth analysis of the Co 2p and Ni 2p spectra. The Co 2p fitting results illustrated in Fig. 3h reveal that the sample contains two sets of spin-orbit doublets (Co $2p_{3/2}$ and Co $2p_{1/2}$) and the corresponding satellite peaks. The 0.6 eV positive shift in Co 2p binding energy stems from increased nucleus and the extranuclear electrons after the metal loses electrons [30]. Compared with CS@NCL/NF, F-CS@NCL/NF exhibits a markedly increased proportion of Co³⁺ alongside a decreased Co²⁺ ratio. These results substantiate that F atoms, which possess a high electronegativity (3.98), effectively elevate the oxidation states of the metal elements by inducing electron transfer [27]. Concurrently, the Ni 2p spectrum manifests a synchronized modulation trend. the characteristic peaks of Ni²⁺ and Ni³⁺ exhibit a negative shift of 0.4 eV (Fig. 3g). The observed spectral shift is ascribed to the accumulation of electrons on the Ni centers. Based on the work function analysis (Fig. 3k), the built-in electric field drive electrons from the CoS/NF domain to the NCL/NF domain. Consequently, Ni acts as an electron acceptor, leading to an increased electron cloud density and a reduced binding energy. The S 2p spectrum (Fig. 3i) further corroborates the sophisticated regulation of the electronic structure. Additionally, the peak at 168.8 eV is ascribed to Co–S–O species, stemming from the inevitable surface chemical oxidation of the starting materials during synthesis or storage processes [20]. Relative to CS@NCL/NF, the S^{2–} signals of F-CS@NCL/NF exhibit positive shifts of 0.2 eV ($2p_{1/2}$) and 0.4 eV ($2p_{3/2}$), which reflects a reduction in the electron cloud density at the S sites. This charge redistribution process is driven by the electronegative induction of F atoms withdraw electrons from Co, subsequently driving electron migration from S to Co to compensate for the deficiency [27]. Such electron flow, stimulated by the coupling between F-doping and the heterointerface, not only diminishes the electron density of S but also accelerating the interfacial reaction kinetics by optimizing the Co coordination environment [14].

3.2. Electrochemical performance analysis

To evaluate their charge storage behaviors, these synthesized samples were examined utilizing a standard three-terminal setup. As depicted by the CV profiles (Fig. 4a), the F-CS@NCL/NF electrode exhibits an enclosed region noticeably bigger than others, which aligns perfectly with its outstanding specific capacity. The GCD profiles of the four electrodes (Fig. 4b) reveal F-CS@NCL/NF exhibits the longest discharge duration to signify its superior specific capacity. This finding aligns with the trend observed in the CV curves and provides robust evidence that the synergistic effect between the dual heterogeneous interfaces significantly enhances rate performance by optimizing charge transfer kinetics. Fig. 4d displays the CV curves of F-CS@NCL/NF between 5 and 50 mV s⁻¹ where the consistency of the curve profiles across various scan rates indicates outstanding ion transport capability and favorable electrochemical reversibility. The agile kinetic response at high scan rates aligns with the IEF mechanism established by earlier M-S and work function analyses. Induced by the Fermi level difference, this IEF provides an additional driving force to diminish the charge transfer barrier and accelerate reactions. Furthermore, as scan speeds increase, redox peak potentials show minor displacements. This phenomenon stems from delayed charge transport dynamics caused by ohmic, ionic diffusion, and polarization resistances [31]. Finally, the F-CS@NCL/NF CV curves display two distinct redox pairs, whose specific electrochemical mechanisms are clarified by equations 6–11 [5,14].

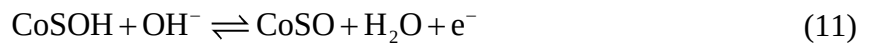
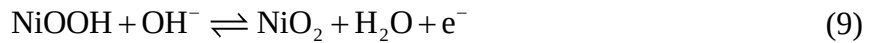
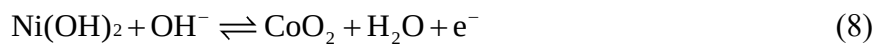
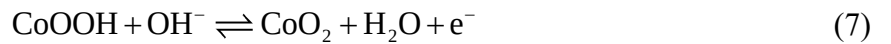
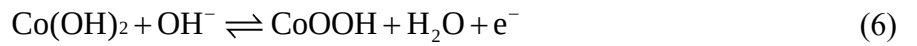


Fig. 4e illustrates the GCD profiles of the F-CS@NCL/NF electrode at diverse current densities, and the derived specific capacities at 1, 2, 3, 5 and 10 A g⁻¹ are 3442, 3260 3205, 3112 and 3025F g⁻¹ respectively, confirming the superior charge storage capability of F-CS@NCL/NF. Upon scaling up the current density from 1 to 10 A g⁻¹, the GCD curves of F-CS@NCL/NF remain well-preserved in shape to reflect enhanced electrochemical reversibility and outstanding rate capability [32]. To identify the optimal F-doping concentration, a gradient experiment was performed using ammonium fluoride (Fig. 4f) where systematic characterization revealed that the sample prepared with 1 M

concentration achieved the longest GCD discharge duration and best performance, leading to the selection of 1 M as the optimal doping concentration. Additionally, Fig. 4i illustrates despite the increased current density reaching 10 A g^{-1} , F-CS@NCL/NF retains 87.8% of its initial capacity, significantly outperforming NCL/NF (53.7%), CoS/NF (53.3%), and CS@NCL/NF (73.5%), thereby confirming that F-doping effectively alleviates the space charge transfer resistance to facilitate superior rate performance.

To gain profound insights into the reaction kinetics and ion transport behavior at the electrode interface, EIS measurements were systematically conducted across the series of samples. The Nyquist plots reveal that the F-CS@NCL/NF electrode manifests the lowest charge transfer resistance (R_{ct}) value of 0.87Ω . This result is particularly noteworthy when compared to the higher resistances observed for NCL/NF (1.03Ω), CoS/NF (1.11Ω), and CS@NCL/NF (0.94Ω), thereby underscoring the superior conductivity of the F-CS@NCL/NF material. These findings provide robust evidence for the elevated carrier concentration reported in the prior M-S analysis, substantiating that the synergy between F-doping and heterojunction formation successfully diminishes interfacial energy barriers. Such an architecture ultimately facilitates a marked acceleration in the kinetic efficiency of Faradaic reaction.

The synergistic effect between F-doping and the p-p heterojunction similarly endows the electrode with outstanding cycling durability. F-CS@NCL/NF exhibits a mere 11.19% capacity attenuation after extensive cycling (Fig. 4g), while scanning electron microscopy images obtained after 10,000 GCD cycles demonstrate that the material maintains its original morphology, which underscores superior structural stability of the electrode. This superior performance is derived from the deep coupling between the well-designed surface-decorated heterostructure and F-doping. F-doping effectively suppresses lattice distortion through the formation of strong polar bonds and introduces stable defect sites to maintain structural integrity during long-term cycling, thereby successfully preventing the agglomeration and collapse of the active species.

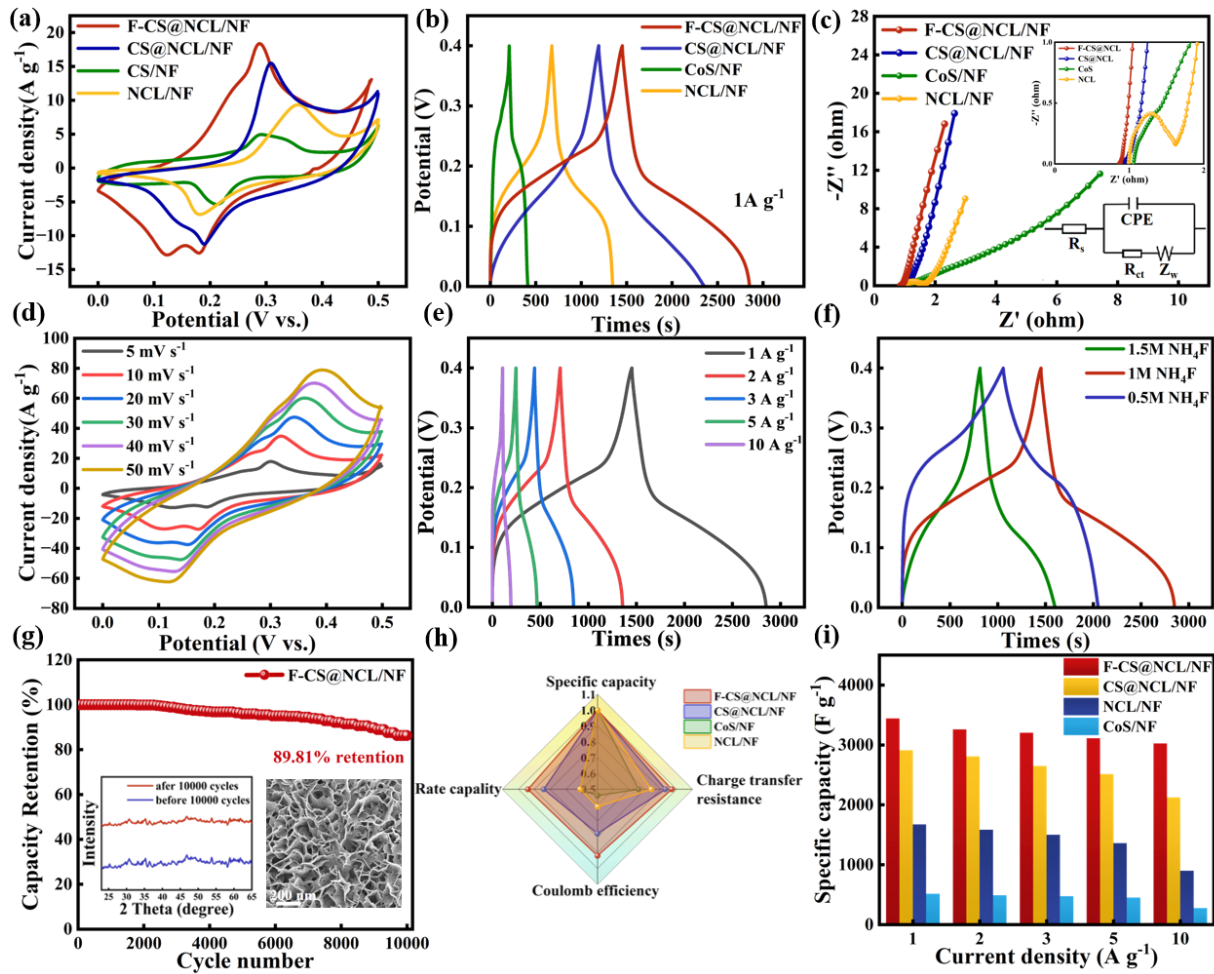


Fig. 4. (a) CV profiles comparing various electrodes at 5 mV s⁻¹. (b) GCD curves of the respective electrodes at 1 A g⁻¹. (c) Nyquist plots of the four electrodes. (d) CV curves for F-CS@NCL/NF across scan rates from 5 to 50 mV s⁻¹. (e) GCD curves of the F-CS@NCL/NF across different current densities. (f) Effect of Different Concentrations of Ammonium Fluoride on GCD Performance During F-doping. (g) Cycling stability of the F-CS@NCL/NF at 10 A g⁻¹. (h) Specific capacity of the four electrodes as a function of current densities. (i) Specific capacity of the four electrodes as a function of current densities.

Fig. 4h compares the electrochemical performance of different electrodes from four dimensions, including coulombic efficiency, specific capacity, rate capability, and charge transfer resistance. It can be clearly observed that the F-CS@NCL/NF electrode exhibits an overall performance advantage over the other reference electrodes in all these parameters. This performance enhancement is closely associated with the introduction of F-doping. By modulating the band structure and charge distribution, doping significantly boosts conductivity. Consequently, the outstanding electrochemical behavior of F-CS@NCL/NF, outperforming its three counterparts, originates from the synergy

between F-doping and the p-p heterojunction.

3.3. Electrode Reaction Kinetics

To further elucidate the charge storage kinetics regarding NCL/NF, CoS/NF, CS@NCL/NF, and F-CS@NCL/NF, CV curves were collected across recorded at diverse scan rates using a three-electrode configuration. The correlation between the peak current and the scan rate at a specific potential is described by Equation 12 [33].

$$i = av^b \quad (12)$$

Therein, a and b signify the fitting constants, while i and v represent the peak current and scan velocity. The b value typically corresponds to the slope of the $\log(i)$ versus $\log(v)$ fitting curve and serves as a critical physical indicator for determining the energy storage mechanism [8]. Calculated b values (0.59–0.78, Fig. 5) indicate a hybrid storage mechanism dominated by capacitive contribution. The experimental anodic and cathodic b values for the four materials are determined to be 0.62 and 0.71 (Fig. 5j), 0.74 and 0.78 (Fig. 5g), 0.62 and 0.63 (Fig. 5d), and 0.59 and 0.63 (Fig. 5a), respectively. These results demonstrate that the energy storage processes in all samples are governed by the synergistic interaction of diffusion-controlled and capacitive-controlled mechanisms. Within this context, the charge storage behavior of F-CS@NCL/NF is characterized by a complex coupling of both modes and is primarily influenced by capacitive control. Notably, the b value for F-CS@NCL/NF is slightly lower than that of CS@NCL/NF, an evolution that reflects an enhanced contribution from diffusion control. This shift confirms that F-doping effectively enhances Faradaic reaction kinetics by refining ion transport pathways.

The current was deconvoluted using Equation 13 to evaluate the respective fractions of capacitive and diffusion-limited processes. In this relationship, k_1v represents the current component arising from capacitive control while $k_2v^{1/2}$ corresponds to the contribution from diffusion-controlled mechanisms [34].

$$i(V) = k_1v + k_2v^{1/2} \quad (13)$$

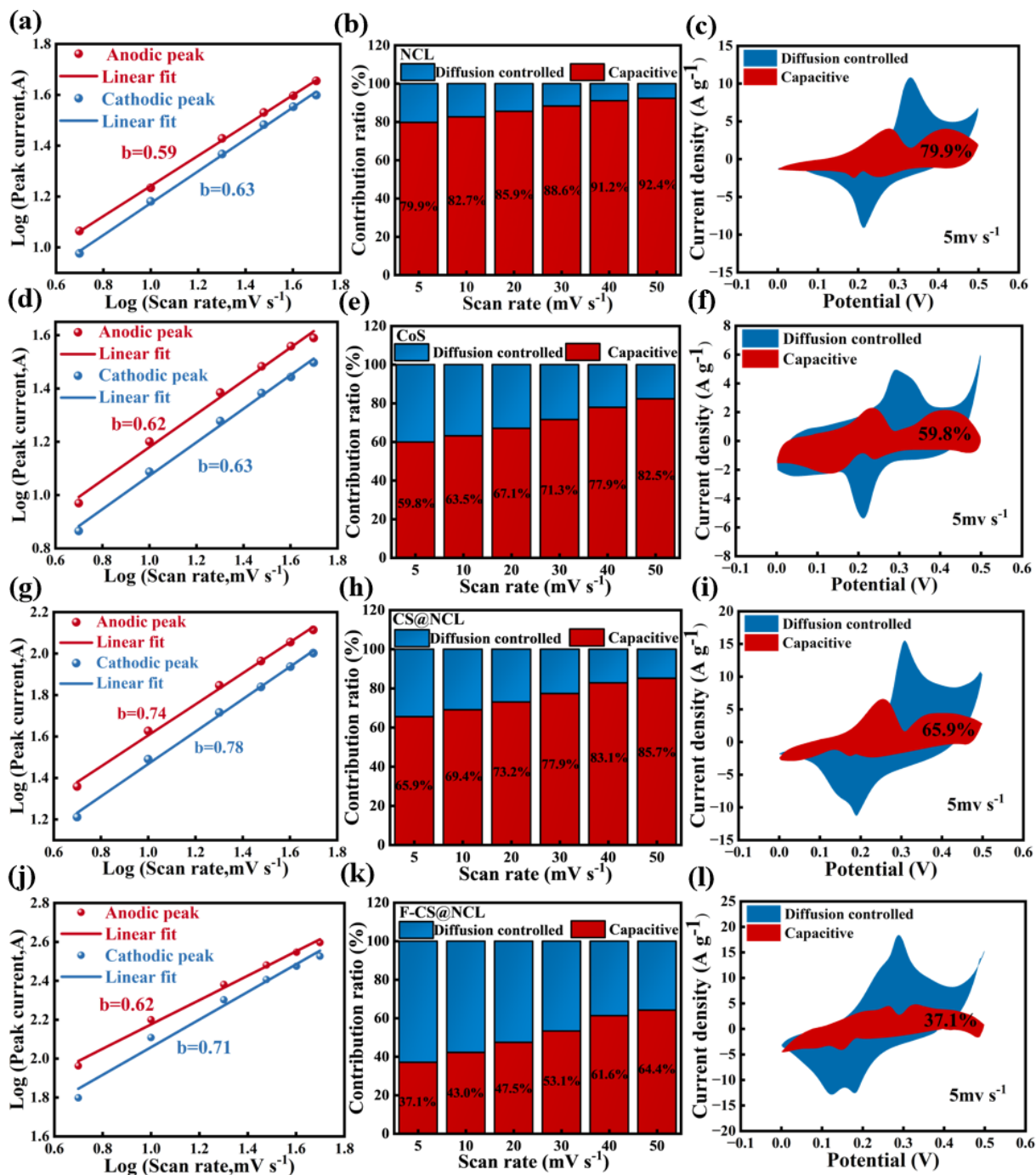


Fig. 5. (a, d, g, j) Linear fitting of $\log(i)$ versus $\log(v)$ for NCL/NF, CoS/NF, CS@NCL/NF, and F-CS@NCL/NF. (b, e, h, k) The respective contribution of the diffusion controlled and capacitive processes in total charge storage at different scan rates. (c, f, i, l) The corresponding CV curves showing the capacitive contribution and the diffuse controlled contribution at 5 mV s^{-1} .

The capacitive contributions for NCL/NF, CoS/NF, CS@NCL/NF, and F-CS@NCL/NF (Fig. 5c, Fig. 5f, Fig. 5i and Fig. 5l) at a scan rate of 5 mV s^{-1} are 79.9%, 59.8%, 65.9%, and 37.1%, respectively. Subsequent analysis reveals that the proportion of capacitive control for each sample

increases alongside the elevation of the scan rate, a trend that aligns with the characteristics of capacitive energy storage where rapid surface charge transfer processes become more dominant under rapid scanning conditions.

3.4. Density functional theory calculation

DFT calculations were employed to uncover the influence of F-doping and the heterointerface on electronic structures and kinetics. Fig. 6a compares the total DOS for NCL, CoS, CS@NCL, and F-CS@NCL, where the results indicate that the electronic synergy between NCL and CoS significantly enhances the energy level distribution of CS@NCL near the Fermi level, strongly substantiating that the interfacial effect of the p-p heterojunction can further improve the electronic conductivity of the material. Crucially, F-CS@NCL displays a higher energy level density near the Fermi level than CS@NCL, signifying that the interfacial coupling effect of F-doping has effectively accelerated the charge transfer process. Fig. 6c presents the partial density of states (PDOS) of the electrode materials. Based on the d band center (ε_d) theory, the position of the ε_d for transition metal elements is intrinsically linked to their adsorption capacity for the medium. Within the established theoretical framework, a closer ε_d to the Fermi level generally leads to a more prominent electron-electrode interaction [35]. Compared with the values of CS@NCL (−0.846 eV), CoS (−1.072 eV), and NCL (−1.311 eV), the ε_d of F-CS@NCL (−0.594 eV) is located nearest to the Fermi level. This proximity demonstrates that the synergy between surface F-doping and the p-p heterojunction enhances the adsorption capacity of the electrode material for OH[−] during the electrochemical reaction.

This work performed a detailed calculation of the differential charge density (Fig. 6b) with the objective of profoundly clarifying the charge transfer magnitude and spatial distribution of F-CS@NCL. Within this visualization, the blue and yellow regions represent the electron dissipation zones and accumulation respectively, whereby a clear aggregation of electrons is observed at the NCL/CoS hetero-interface. This is attributed to the interfacial interaction generated by the combination of the two components, which can effectively drive the redistribution of interfacial charges and thus significantly increase the interfacial charge density. Furthermore, the presence of distinct charge accumulation and dissipation surrounding the F-doping sites signifies the dual functionality of F-doping in promoting charge carrier transport and improving conductivity while simultaneously strengthening the coupling between the interfaces. To evaluate the thermodynamic feasibility F-doping and identify the optimal doping site, this work systematically calculated the

formation energies of fluorine atoms at the Ni and Co sites of NCL as well as the Co sites within the CoS shell. The formation energy of F–CoS–Co reaches a minimum value of -2.37 eV (Fig. 6d) which implies that fluorine atoms preferentially occupy the exposed cobalt sites in the CoS shell. This thermodynamically favorable site exhibits high consistency with the experimental synthesis route to ensure a precise match between the theoretical model and the structural characteristics of the experimental samples. Additionally, the CS@NCL heterojunction demonstrates an exceptionally low formation energy of -5285.03 eV as detailed in Fig. S5. This result signifies that the surface-decorated architecture possesses superior stability and can form spontaneously to provide a reliable structural foundation for the subsequent F-doping process.

Dictated by the Sabatier principle, H_2O adsorption energy crucially regulates the overall electrochemical kinetics. Specifically, within alkaline systems, water molecules must undergo dissociation to generate H^* active intermediates whereby the adsorption strength of this process not only directly determines the adsorption and desorption kinetics of reaction intermediates but also profoundly influences the electrochemical reaction energy barriers and the overall kinetic efficiency [36]. The specific values of water adsorption energy (ΔE_{H_2O}) were calculated in accordance with Equation 14 [37].

$$\Delta E_{H_2O} = E_{\text{slab} + H_2O} - E_{\text{slab}} - E_{H_2O} \quad (14)$$

The DFT calculation results presented in Fig. 6e reveal that the ΔE_{H_2O} values for NCL, CoS, CS@NCL, and F–CS@NCL are -0.48 , -0.59 , -0.78 and -1.02 eV respectively. Compared with the other control samples, F–CS@NCL exhibits more superior thermodynamic adsorption behavior toward water molecules. This optimized adsorption strength assists in the initial adsorption and activation of reactant molecules on the electrode surface while simultaneously promoting the efficient desorption of reaction intermediates to provide critical thermodynamic support for the efficient and continuous progression of the electrochemical reaction. It is noteworthy that the absolute value of ΔE_{H_2O} for F–CS@NCL is significantly higher than those of the other samples. This phenomenon indirectly indicates that its surface may possess more abundant active sites which exerts a prominent positive effect on accelerating the electrochemical reaction rate and strengthening the charge transfer kinetics at the electrode–electrolyte interface.

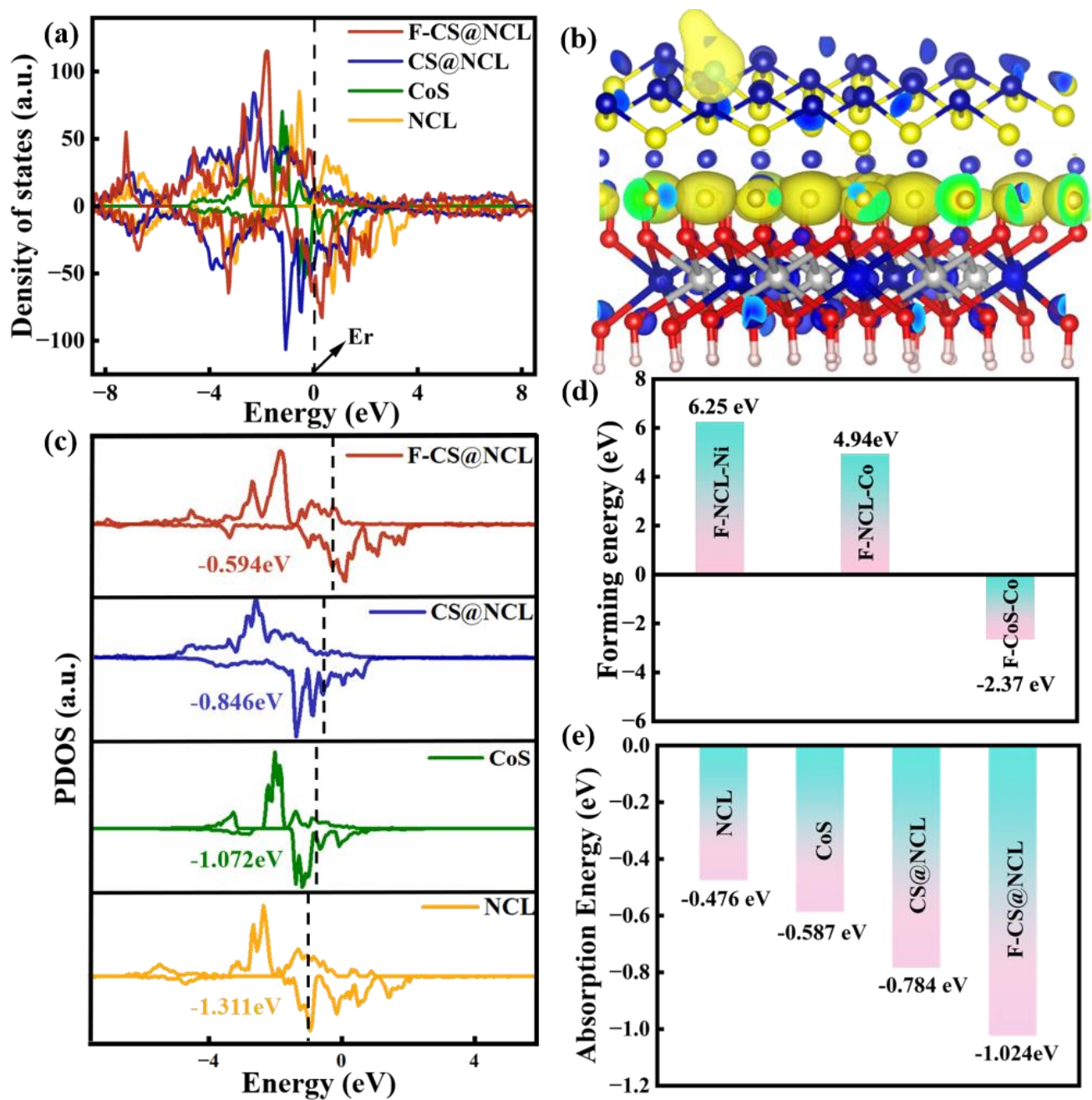


Fig. 6. (a) DOS of the NCL, CoS, CS@NCL and F-CS@NCL. (b) The differential density charge map of CS@NCL and F-CS@NCL. (c) PDOS of the NCL, CoS, CS@NCL and F-CS@NCL. (d) Formation energy of F-NCL-Co, F-NCL-Ni and F-CoS-Co. (e) H₂O adsorption energy of the NCL, CoS, CS@NCL and F-CS@NCL.

3.5. Electrochemical Analysis of F-CS@NCL/NF//AC Hybrid Supercapacitor

To investigate the electrochemical performance of F-CS@NCL/NF, an HSC device was assembled. This assembly integrated a KOH electrolyte and an ion-exchange membrane separator, with the schematic representation of the device architecture provided in Fig. 7a. Prior to the formal assembly of the HSC, the electrochemical behavior of the AC electrode was systematically evaluated in a three-electrode setup. The CV curves of both AC and F-CS@NCL/NF at a scan rate of 5 mV s^{-1} (Fig. 7b), where they demonstrate reliable stability within the potential ranges of -1 to 0 V and 0 to 0.5 V , respectively. This compatibility establishes the physical foundation for expanding the operating voltage window of the F-CS@NCL/NF //AC device to 1.5 V . The CV curves for the AC electrode within a potential range of -1 to 0 V (Fig. 7c) vs Hg/HgO at various scan rates. The curves maintain a highly consistent approximately rectangular shape without any redox peaks. This behavior reveals that the AC electrode's storage mechanism is dominated by interfacial EDLC [38].

Table 1. Electrochemical properties of F-CS@NCL//AC HSC compared with other SCs.

Type of SCs	Current density	Specific capacitance	Power density [W kg ⁻¹]	Energy density [W h kg ⁻¹]	Coulombic efficiency	Cycle performance	Refs.
NiCo-LDH@TCDs3.0	1 A/g	122.4 F g ⁻¹	800	87	99%	70% (5000)	[1]
CFP@Cedldh//AC	1 A/g	248 F g ⁻¹	800	90	99%	86.5% (10000)	[2]
NiFe-LDH@CoS ₂ @Ni	0.5A/g	51.4 F g ⁻¹	375.16	15.84	95%	/	[4]
NiCo-LDH/MXene	1 A/g	271 F g ⁻¹	699.1	73.8	97%	83.1% (10000)	[5]
NiCo-LDH-MoO ₄ ²⁻	1 A/g	66.2 F g ⁻¹	800	37.1	98%	90.2% (10000)	[6]
MnO ₂ @NiCoLDH/CoS ₂ //AC	1 A/g	144.7F g ⁻¹	391.5	49.5	98%	51.0% (2000)	[25]
NNCLDH/OPC	20 A/g	141 F g ⁻¹	812	51	98%	73.3% (4000)	[29]
MIL@CuFe-Se//AC	1 A/g	219 F g ⁻¹	800	90	99.5%	83.4% (10000)	[32]
Ni CoP @CoS//AC	2A/g	179.6 F g ⁻¹	748.9	35.8	91%	91.4% (5000)	[39]
Ni Co P/C//AC	1 A/g	171.9 F g ⁻¹	750	53.7	98%	81.3% (10000)	[40]
Ni Co-LDH/r GO	0.5 A/g	159.6 F g ⁻¹	375.2	49.9	94%	62.2% (10000)	[41]
Cu O@ Ni Co-LDH//AC	0.2 A/g	191.4F g ⁻¹	556.3	20.1	95%	87.1% (6000)	[42]
Ni Co LDH/C ₃ N ₄	1 A/g	154F g ⁻¹	740	33	96%	97% (9000)	[43]
NCLP@NiMn-LDH	0.5 A/g	155 F g ⁻¹	750	42.2	95%	80% (10000)	[44]
CoSe ₂ /C//RGO	1 A/g	57 F g ⁻¹	699	21	94%	97% (10000)	[45]
F-CS@NCL//AC	1 A/g	364.3 F g ⁻¹	750	114.6	99.8%	86.3% (10000)	This Work

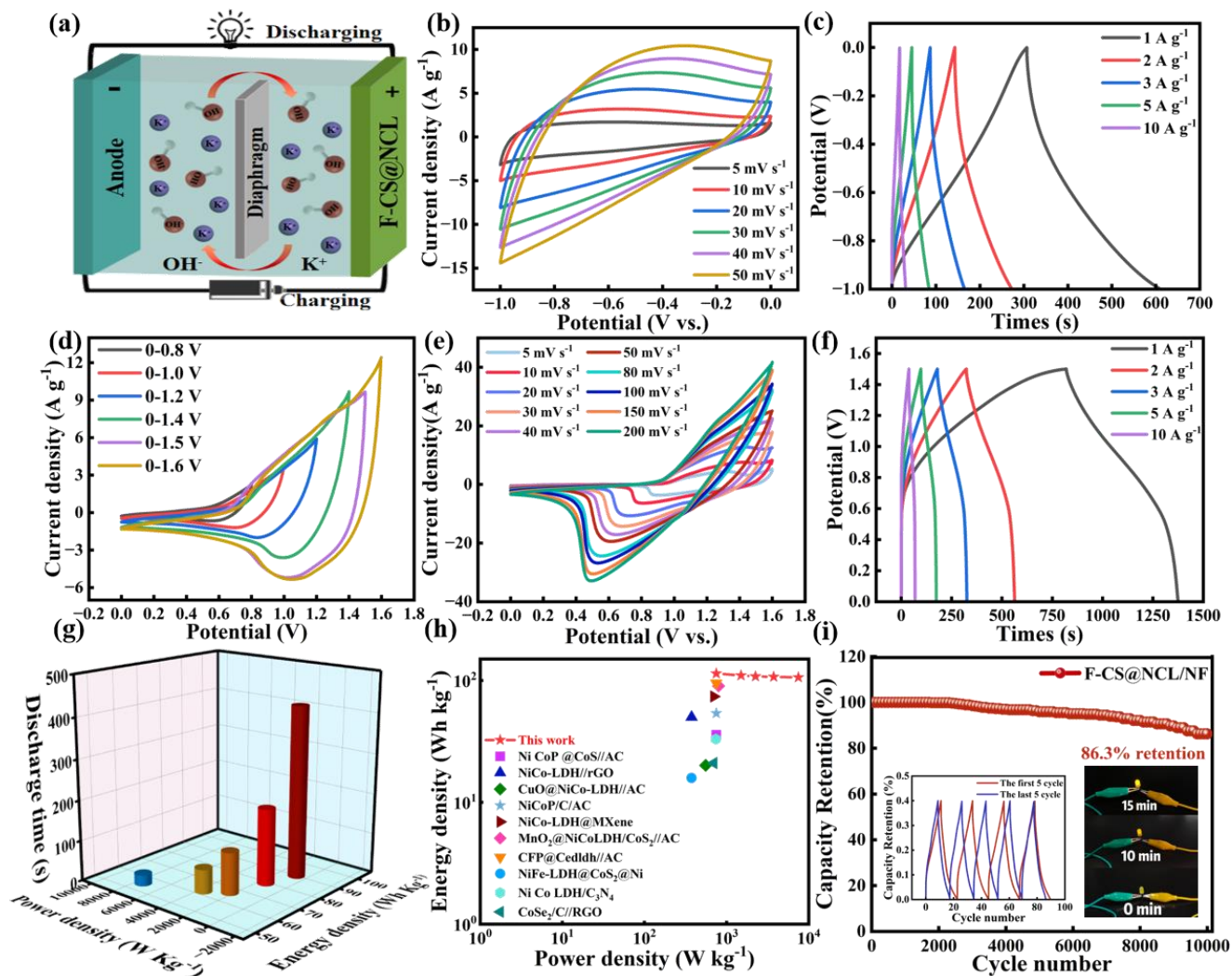


Fig. 7. (a) Energy storage mechanism of F-CS@NCL//AC HSC. (b) CV curves for the AC at multiple scan rates. (c) GCD responses of the AC under varied current densities. (d) CV curves for the HSC device across diverse voltage windows. (e) CV curves for the HSC within the potential range of 0–1.5 V. (f) GCD curves of the HSC at different current densities. (g) The relationship diagram of the power density, energy density and discharge time of the HSC equipment. (h) Ragone plots of the F-CS@NCL//AC HSC device and other SCs recorded in references. (i) Cycling stability and coulombic efficiency of the HSC equipment at 10 A g⁻¹.

To determine the most effective operating range for the F-CS@NCL//AC HSC, CV curves were recorded across different voltage windows (Fig. 7d). Increasing the voltage to 1.6 V leads to noticeable polarization, which signals a reduction in electrochemical stability and confirms that the optimal range for the device is 0 to 1.5 V. This successful expansion of the voltage window originates from the synergy of multiple factors. The stable potential window of F-CS@NCL/NF

optimize charge storage mechanisms while simultaneously improving energy and power densities. The redistribution of charge at the interface between the positive and negative materials also helps regulate the local electric field, providing a foundation for stable operation across a broad potential range. Furthermore, the CV curves maintained a similar shape with well-preserved redox peaks demonstrating superior rate performance even at an accelerated scan rate of 200 mV s^{-1} and fast ion transport kinetics (Fig. 7e). At 1, 2, 3, 5, and 10 A g^{-1} , the device delivers corresponding specific capacitances of 364.3, 308.1, 285.3, 245.1, and 215.4 F g^{-1} (Fig. 7f).

After 10,000 charge-discharge cycles, the device experiences merely a 13.7% capacitance decay, verifying its exceptional long-term durability. The profiles of the GCD curves before and after this extensive cycling show no significant differences (Fig. 7i). The HSC device also shows an impressive match between energy and power density, reaching 114 W h kg^{-1} at a power density of 750 W kg^{-1} (Fig. 7g). The comprehensive results presented in the Ragone plot in Fig. 7h and the comparison in Table 1 demonstrate that this device outperforms many previously reported systems [39–45]. Two series-connected F-CS@NCL // AC HSCs can power a light bulb for 15 min. These findings collectively suggest that the F-CS@NCL//AC based hybrid supercapacitor offers superior energy storage properties and showcases immense prospects for future commercial utilization in high-performance energy storage technologies.

4. Conclusion

In summary, through a dual-step electrodeposition and subsequent soaking-doping strategy, we successfully construct F-CS@NCL/NF composite electrodes, benefiting from the synergistic regulation of p-p heterojunctions and F-doping. Multiple characterization results combined with DFT calculations confirm that the p-p heterojunction formed between CoS/NF and NCL/NF can achieve directional regulation of charges by virtue of the difference in work function, thus effectively driving the rearrangement of interfacial charges and improving the charge transfer rate. Concurrently, F-doping atoms preferentially occupy the cobalt sites of the CoS/NF shell to significantly strengthen the adsorption capacity of the material for OH^- by further optimizing the electronic structure and narrowing the bandgap. This synergistic modification effect successfully overcomes the performance bottlenecks of poor electrical conductivity and slow ion diffusion inherent in single components. In practical application demonstrations, the series-connected device assembled from this material can drive an LED to emit light continuously for 15 mins, which fully highlights its exceptional energy

storage potential. This synergistic strategy not only offers a fresh perspective for designing high-performance supercapacitors but also provides valuable insights for interface engineering in broader electrochemical fields.

CRedit authorship contribution statement

Jibo Jiang: Writing – review & editing, Supervision, Funding acquisition. **Zihan Zhang:** Writing – review & editing, Writing – original draft, Software, Resources, Investigation, Formal analysis. **Qianqian Shan:** Validation, Project administration, Investigation, Formal analysis. **Kairan Hu:** Writing – review & editing, Validation. **Jie Yang:** Validation, Supervision. **Sheng Han:** Supervision, Funding acquisition.

Declaration of Interest Statement

There is no conflict of interest in the submission of this manuscript, and the manuscript has been approved for publication by all authors.

Acknowledgements

This work was supported by the following funding sources: the National Natural Science Foundation of China (Grant No. 22278269), the Joint Fund of the Ministry of Education (Grant No. 8091B02052304) and the Scientific Research Project of Shanghai Municipal Education Commission (Grant No. 2023ZKZD54).

References

- [1] S. Wang, J. Deng, M. Li, et al., Nitrogen-doped tannin carbon dots anchored NiCo-LDH composites for high-performance asymmetric supercapacitors, *Chem. Eng. J.* 511 (2025) 162275. <http://dx.doi.org/10.1016/j.cej.2025.162275>.
- [2] Z. Zhang, S. Wang, K. Hu, et al., Introducing Ce into CuFePBA@CuCo-LDH to construct Mott-Schottky heterojunctions with optimized electron redistribution for high-performance supercapacitors, *Adv. Funct. Mater.* 35 (2025) 2502025. <https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.202502025>.
- [3] W.Y. Chen, T.T. Wei, L.E. Mo, et al., CoS₂ nanosheets on carbon cloth for flexible all-solid-state supercapacitors, *Chem. Eng. J.* 400 (2020) 125856. <http://dx.doi.org/10.1016/j.cej.2020.125856>.
- [4] J. Tian, A. Zhang, R. Liu, et al., Preparation of CoS₂ supported flower-like NiFe layered double hydroxides nanospheres for high-performance supercapacitors, *J. Colloid Interface*

Sci. 579 (2020) 607–618. <http://dx.doi.org/10.1016/j.jcis.2020.06.086>.

- [5] M. Sun, Y. Wang, J. Sunarso, et al., Synthesis and characterization of two-faced brush-like MXene anchored NiCo-LDH electrode for high-performance supercapacitors, *Appl Energy* 361 (2024) 122944. <http://dx.doi.org/10.1016/j.apenergy.2024.122944>.
- [6] M. Shi, L. Wang, D. Zhang, Implementation of durable structural supercapacitors with molybdate-ion-intercalated NiCo-LDH and polymer-cement composite, *COMPOS PART B-ENGINE* 295 (2025) 112209. <https://doi.org/10.1016/j.compositesb.2025.112209>.
- [7] X. Huang, R. Sun, Y. Li, et al., Two-step electrodeposition synthesis of heterogeneous NiCo-layered double hydroxides@MoO₃ nanocomposites on nickel foam with high performance for hybrid supercapacitors, *Electrochim. Acta* 403 (2022) 139680. <https://linkinghub.elsevier.com/retrieve/pii/S0013468621019642>.
- [8] Q. Jia, H. Li, S. Lu, et al., Thermodynamic mechanisms of Co-S bond anchoring in few-layered 1T-MoS₂ for enhanced capacitive performance via spin state regulation and ion diffusion kinetics, *Energy Environ. Mater* (2026) e70218. <https://onlinelibrary.wiley.com/doi/10.1002/eem2.70218>.
- [9] H. Hu, B.Y. Guan, X.W.D. Lou, Construction of complex CoS hollow structures with enhanced electrochemical properties for hybrid supercapacitors, *CHEM-US* 1 (2016) 102–113. <http://dx.doi.org/10.1016/j.chempr.2016.06.001>.
- [10] F. Song, J. Ma, G. Wang, et al., Novel rough nanorods NiO-(NiFeCo)O as positive electrode of high-performance asymmetric supercapacitors assembled with SiO₂/rGO negative electrode, *Energy* 326 (2025) 136360. <http://dx.doi.org/10.1016/j.energy.2025.136360>.
- [11] Y. Zhao, Y. Zhu, C. Xi, et al., Manipulating NiS₂/MXene-induced Mott-Schottky heterostructures via precursor doping strategy for promoting alkaline electrocatalytic overall water splitting, *APPL CATAL B-ENVIRON* 378 (2025) 125611. <https://doi.org/10.1016/j.apcatb.2025.125611>.
- [12] C. Chen, Y. Wang, J. Li, et al., In situ construction of heteroatom F-doped Mn₃O₄ spinel catalysts with robust activity and SO₂ resistance for NH₃-SCR at low temperature, *Appl. Catal. B: Environ.* 338 (2023) 123086. <https://doi.org/10.1016/j.apcatb.2023.123086>.
- [13] X. Wang, J. Li, Q. Wang, et al., Balancing electron spin states of Na_{3.12}Fe_{2.44}(P₂O₇)₂thro

ugh F-doping realized high performance on sodium ions storage, *Adv. Energy Mater* (2025). <https://doi.org/10.1002/aenm.202502300>.

- [14] X. Chen, Z. Zhang, S. Zhou, et al., In-situ growth transformation and oxygen vacancy synergistic modulation of the electronic structure of NiCo-LDH enables high-performance hybrid supercapacitors, *Appl Energy* 371 (2024) 123670. <http://doi.org/10.1016/j.apenergy.2024.123670>.
- [15] Y. Huang, X. Zhang, L. Li, et al., Mott-Schottky barrier enabling high-performance hydrazine-assisted hydrogen generation at ampere-level current densities, *Adv. Funct. Mater* (2024). <http://dx.doi.org/10.1002/adfm.202401011>.
- [16] Q. Han, W. Yang, W. Li, et al., MXene-based separators for redox-enhanced electric capacitors with a suppressed shuttle effect and self-discharge: the effect of MXene ageing, *New J. Chem* 47 (2023) 3516–3523. <http://dx.doi.org/10.1039/d2nj05439f>.
- [17] R. Nasser, H. Zhou, H. Elhouichet, et al., Constructing Mott-Schottky heterojunctions in transition metal compounds for enhanced electrocatalytic water splitting performance, *Chem. Eng. J* 489 (2024) 151554. <http://dx.doi.org/10.1016/j.cej.2024.151554>.
- [18] S. Ni, H. Qu, Z. Xu, et al., Interfacial engineering of the NiSe₂/FeSe₂ p-p heterojunction for promoting oxygen evolution reaction and electrocatalytic urea oxidation, *Appl. Catal. B: Environ.* 299 (2021) 120638. <https://doi.org/10.1016/j.apcatb.2021.120638>.
- [19] X. Han, J. Li, J. Lu, et al., High mass-loading NiCo-LDH nanosheet arrays grown on carbon cloth by electrodeposition for excellent electrochemical energy storage, *Nano Energy* 86 (2021) 106079. <http://dx.doi.org/10.1016/j.nanoen.2021.106079>.
- [20] C.K. Maity, S. De, A. Panigrahi, et al., Aerosol derived carbon dots decorated boron nitride supported Zn-doped MoS₂ for high performing flexible asymmetric supercapacitor, *Compos. Part B-Eng* 264 (2023) 110887. <https://doi.org/10.1016/j.compositesb.2023.110887>.
- [21] H. Zhang, B. Yan, H. Zhao, et al., Hollow nanostructure boosts the surface capacitive charge storage of NiCo-LDH derived from metal-organic framework for high performance asymmetric supercapacitor, *J. Alloy. Compd* 896 (2022) 163019. <http://dx.doi.org/10.1016/j.jallcom.2021.163019>.
- [22] Q. Pan, F. Zheng, D. Deng, et al., Interlayer spacing regulation of NiCo-LDH nanosheet

ts with ultrahigh specific capacity for battery-type supercapacitors, *ACS Appl. Mater* 13 (2021) 56692–56703. <http://dx.doi.org/10.1021/acsami.1c19320>.

- [23] R.P. Rane, S.S. Shitole, S.P. Varande, et al., Upcycling waste graphite from spent LIBs for fabrication of novel mesoporous carbon and p-GN/BT based supercapacitor, *Appl Energy* 364 (2024) 123220. <http://dx.doi.org/10.1016/j.apenergy.2024.123220>.
- [24] H. Chen, Y. Zhang, H. Yang, et al., Self-discharge of redox electrolyte enhanced supercapacitors based on nanosheet-like CoS₂, *Chem. Eng. J* 494 (2024) 152910. <https://doi.org/10.1016/j.cej.2024.152910>.
- [25] F.H. Wang, Unique MOF-derived hierarchical MnO₂ nanotubes@NiCo LDH/CoS₂ nanocages materials as high performance supercapacitor, *Chem. Mater* (2018) 1–21. <http://dx.doi.org/10.1039/C9TA01951K>.
- [26] G. Hao, T. Zhao, Q. Fang, et al., Trace F-doped Co₃O₄ nanoneedles for enhanced acidic water oxidation activity via promoting OH coverage, *Green Chem* 26 (2024) 9782–9790. <https://doi.org/10.1039/d4gc01895h>.
- [27] X. Zhang, Y. Chen, S. Song, et al., Self-templated fabrication of heterogeneous F doped NiCo-alkoxides/NiCo hydroxides yolk-shelled spheres with enhanced performance for supercapacitors, *Chem. Eng. J* 521 (2025) 166858. <https://doi.org/10.1016/j.cej.2025.166858>.
- [28] J. Jiang, C. Xi, Y. Zhao, et al., MXene-Based hierarchical electron coupling engineering of F-doped NiFe LDH/MOF-74 electrocatalysts for efficient overall water splitting, *ACS Catal* 15 (2025) 2561–2575. <https://doi.org/10.1021/acscatal.4c07798>.
- [29] S. Yang, M. Li, X. He, et al., One-step synthesis of NiCo-LDH@Ni(OH)₂ heterostructure foams on biomass-derived porous carbon for high-performance supercapacitors, *Small* 21 (2025) 2407822. <https://onlinelibrary.wiley.com/doi/10.1002/sml.202407822>.
- [30] H. Kim, K. Min, G. Song, et al., Hollow-structured cobalt sulfide electrocatalyst for alk

aline oxygen evolution reaction: rational tuning of electronic structure using iron and fluorine dual-doping strategy, *J. Colloid Interface Sci* 665 (2024) 922–933. <http://dx.doi.org/10.1016/j.jcis.2024.03.201>.

- [31] D. Ren, X. Li, X. Zhao, et al., Development and evaluation of Zn^{2+} ions hybrid supercapacitor based on Zn_xMnO_2 -CNTs cathode, *Appl Energy* 324 (2022) 119730. <http://dx.doi.org/10.1016/j.apenergy.2022.119730>.
- [32] Z. Zhang, L. Li, X. Chen, et al., Modulating charge distribution of MIL@CuFe-Se by in-situ interface engineering for high performance supercapacitors, *Chem. Eng. J* 496 (2024) 154370. <http://dx.doi.org/10.1016/j.cej.2024.154370>.
- [33] T. Ramachandran, S.S. Sana, K.D. Kumar, et al., Asymmetric supercapacitors: Unlocking the energy storage revolution, *J. Energy Storage* 73 (2023) 109096. <http://dx.doi.org/10.1016/j.est.2023.109096>.
- [34] Z. Lv, Y. Tang, Z. Zhu, et al., Honeycomb-lantern-inspired 3D stretchable supercapacitors with enhanced specific areal capacitance, *Adv. Mater* 30 (2018). <http://dx.doi.org/10.1002/adma.201805468>.
- [35] J. Mu, Z. Guo, Y. Zhao, et al., ZIF-67/rGO/NiPc composite electrode material for high-performance asymmetric supercapacitors, *J MATER SCI-MATER EL* 33 (2022) 17733–17744. <http://dx.doi.org/10.1007/s10854-022-08636-5>.
- [36] Z. Li, L.Y. Zhang, L. Zhang, et al., ZIF-67-derived CoSe/NC composites as anode materials for lithium-ion batteries, *Nanoscale Res. Lett* 14 (2019). <http://dx.doi.org/10.1186/s11671-019-3194-5>.
- [37] K. Hu, Y. Wan, Z. Zhang, et al., Electrochemical reduction-induced oxygen vacancies and in-situ selenization strategies synergistically construct high-performance supercapacitors, *J. Colloid Interface Sci* 688 (2025) 118–131. <https://doi.org/10.1016/j.jcis.2025.02.127>.
- [38] M. Kuang, P. Han, Q. Wang, et al., CuCo hybrid oxides as bifunctional electrocatalyst for efficient water splitting, *Adv. Funct. Mater.* 26 (2016) 8555–8561. <http://dx.doi.org/10.1002/advfm.201600855>.

1002/adfm.201604804.

- [39] Z. Xu, C. Du, H. Yang, et al., NiCoP@CoS tree-like core-shell nanoarrays on nickel foam as battery-type electrodes for supercapacitors, *Chem. Eng. J* 421 (2021) 127871. <http://dx.doi.org/10.1016/j.cej.2020.127871>.
- [40] X. Wang, W. Li, Y. Xu, et al., NiCoP/C composite with hollow sphere as electrodes for high performance supercapacitors, *Electrochim. Acta* 434 (2022) 141313. <http://dx.doi.org/10.1016/j.electacta.2022.141313>.
- [41] L. Zhang, P. Cai, Z. Wei, et al., Synthesis of reduced graphene oxide supported nickel-cobalt-layered double hydroxide nanosheets for supercapacitors, *J. Colloid Interface Sci* 588 (2021) 637–645. <http://dx.doi.org/10.1016/j.jcis.2021.11.056>.
- [42] X. Huang, R. Yang, H. Yin, et al., CuO@NiCo-LDH core-shell structure for flexible fiber-shaped supercapacitor electrode material, *J Energy Storage* 74 (2023) 109319. <https://linkinghub.elsevier.com/retrieve/pii/S2352152X23027172>.
- [43] Y. Tan, Y. Ren, Y. Zhang, C₃N₄ improved nickel-cobalt layered double hydroxide (NiCo LDH) supercapacitive property, *J Energy Storage* 81 (2024) 110367. <http://dx.doi.org/10.1016/j.est.2023.110367>.
- [44] H. Liang, J. Lin, H. Jia, et al., Hierarchical NiCo-LDH/NiCoP@NiMn-LDH hybrid electrodes on carbon cloth for excellent supercapacitors, *J. Mater. Chem. A* 6 (2018) 15040–15046. <http://pubs.rsc.org/en/content/articlepdf/2018/TA/C8TA05065A>.
- [45] X. Wang, C. Qian, T. Zou, et al., Dodecahedral NC-doped CoSe₂ nanoparticles with excellent stability for high-performance flexible solid-state supercapacitors, *J. Electroanal. Chem* 943 (2023) 117612. <http://dx.doi.org/10.1016/j.jelechem.2023.117612>.

F–doping induced interfacial built–in electric field and charge redistribution construction in CoS@NiCoLDH p–p heterojunctions for high–performance supercapacitors

Zihan Zhang, Qianqian Shan, Kairan Hu, Jie Yang, Jibo Jiang*, Sheng Han*

Department of Chemical Engineering and Energy Technology, Shanghai Institute of Technology, Shanghai 201418, China.

*Corresponding authors.

E–mail addresses: jibojiang0506@163.com (J. Jiang), hansheng654321@sina.com (S. Han).

Table 1. Electrochemical properties of F–CS@NCL//AC HSC compared with other SCs.

Type of SCs	Current density	Specific capacitance	Power density[W kg ^{−1}]	Energy density [W h kg ^{−1}]	Coulombic efficiency	Cycle performance	Refs.
NiCo–LDH@TCdS3.0	1 A/g	122.4 F g ^{−1}	800	87	99%	70% (5000)	[1]
CFP@Cedldh//AC	1 A/g	248 F g ^{−1}	800	90	99%	86.5% (10000)	[2]
NiFe–LDH@CoS ₂ @Ni	0.5A/g	51.4 F g ^{−1}	375.16	15.84	95%	/	[4]
NiCo–LDH@MXene	1 A/g	271 F g ^{−1}	699.1	73.8	97%	83.1% (10000)	[5]
NiCo–LDH–MoO ₄ ^{2−}	1 A/g	66.2 F g ^{−1}	800	37.1	98%	90.2% (10000)	[6]
MnO ₂ @NiCoLDH/CoS ₂ //AC	1 A/g	144.7F g ^{−1}	391.5	49.5	98%	51.0% (2000)	[25]
NNCLDH/OPC	20 A/g	141 F g ^{−1}	812	51	98%	73.3% (4000)	[29]
MIL@CuFe–Se//AC	1 A/g	219 F g ^{−1}	800	90	99.5%	83.4% (10000)	[32]
Ni CoP @CoS//AC	2A/g	179.6 F g ^{−1}	748.9	35.8	91%	91.4% (5000)	[39]
Ni Co P/C//AC	1 A/g	171.9 F g ^{−1}	750	53.7	98%	81.3% (10000)	[40]
Ni Co–LDH//r GO	0.5 A/g	159.6 F g ^{−1}	375.2	49.9	94%	62.2% (10000)	[41]
Cu O@ Ni Co–LDH//AC	0.2 A/g	191.4F g ^{−1}	556.3	20.1	95%	87.1% (6000)	[42]
Ni Co LDH/C ₃ N ₄	1 A/g	154F g ^{−1}	740	33	96%	97% (9000)	[43]
NCLP@NiMn–LDH	0.5 A/g	155 F g ^{−1}	750	42.2	95%	80% (10000)	[44]
CoSe ₂ /C//RGO	1 A/g	57 F g ^{−1}	699	21	94%	97% (10000)	[45]
F–CS@NCL//AC	1 A/g	364.3 F g ^{−1}	750	114.6	99.8%	86.3% (10000)	This Work

**F–doping induced interfacial built–in electric field and charge redistribution
construction in CoS@NiCoLDH p–p heterojunctions for high–performance
supercapacitors**

Zihan Zhang, Qianqian Shan, Kairan Hu, Jie Yang, Jibo Jiang*, Sheng Han*

Department of Chemical Engineering and Energy Technology, Shanghai Institute of Technology,
Shanghai 201418, China.

*Corresponding authors.

E–mail addresses: jibojiang0506@163.com (J. Jiang), hansheng654321@sina.com (S. Han).

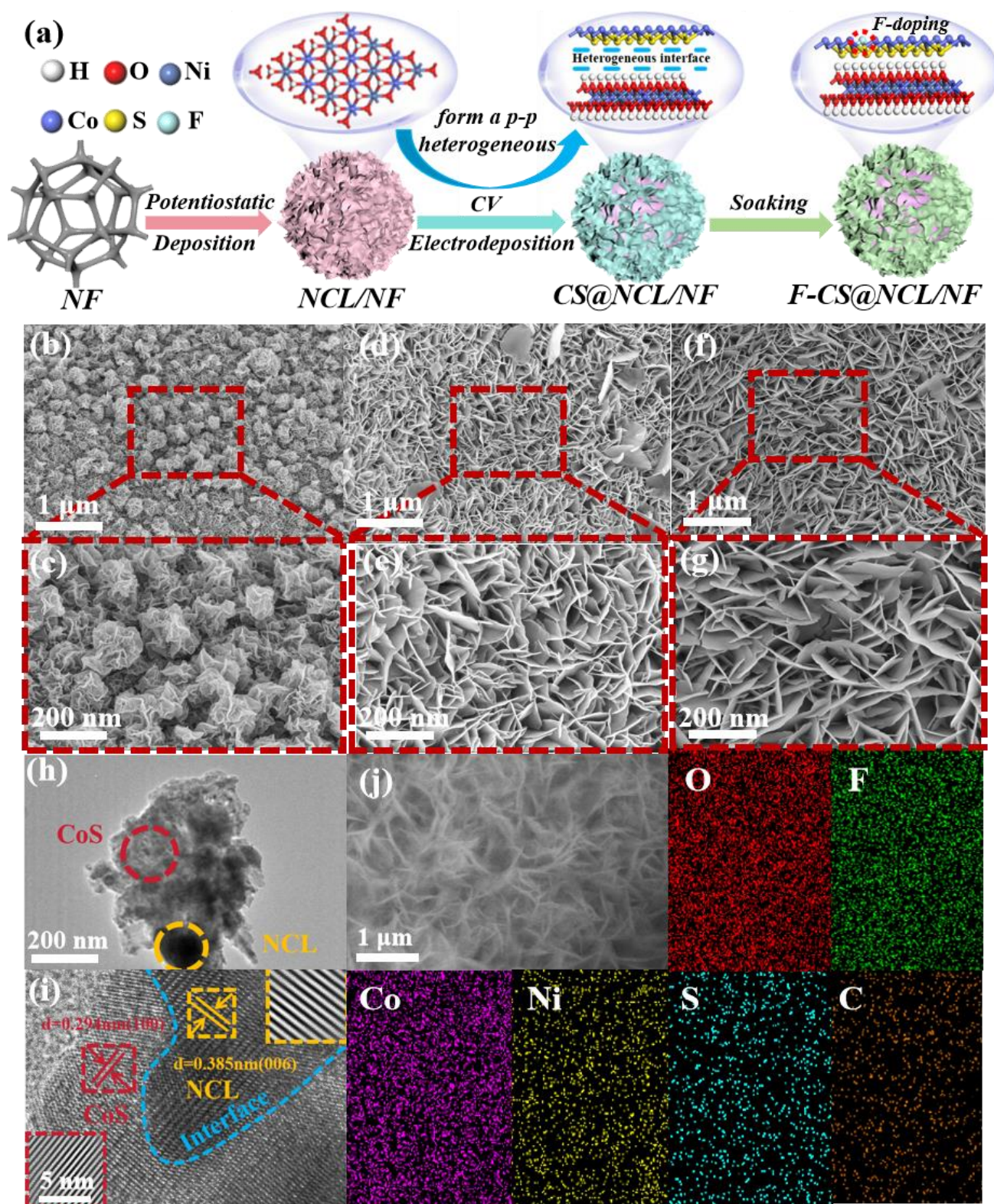


Fig. 1. (a) Schematic diagram depicting the preparation route for F-CS@NCL/NF. SEM micrographs corresponding to (b–c) NCL, (d–e) CS@NCL/NF, and (f–g) F-CS@NCL/NF. (h) TEM images of F-CS@NCL/NF. (i) HRTEM images of F-CS@NCL/NF. (j) HAADF image and corresponding element mapping images of F-CS@NCL/NF. (Repeat each test three times to minimize errors).

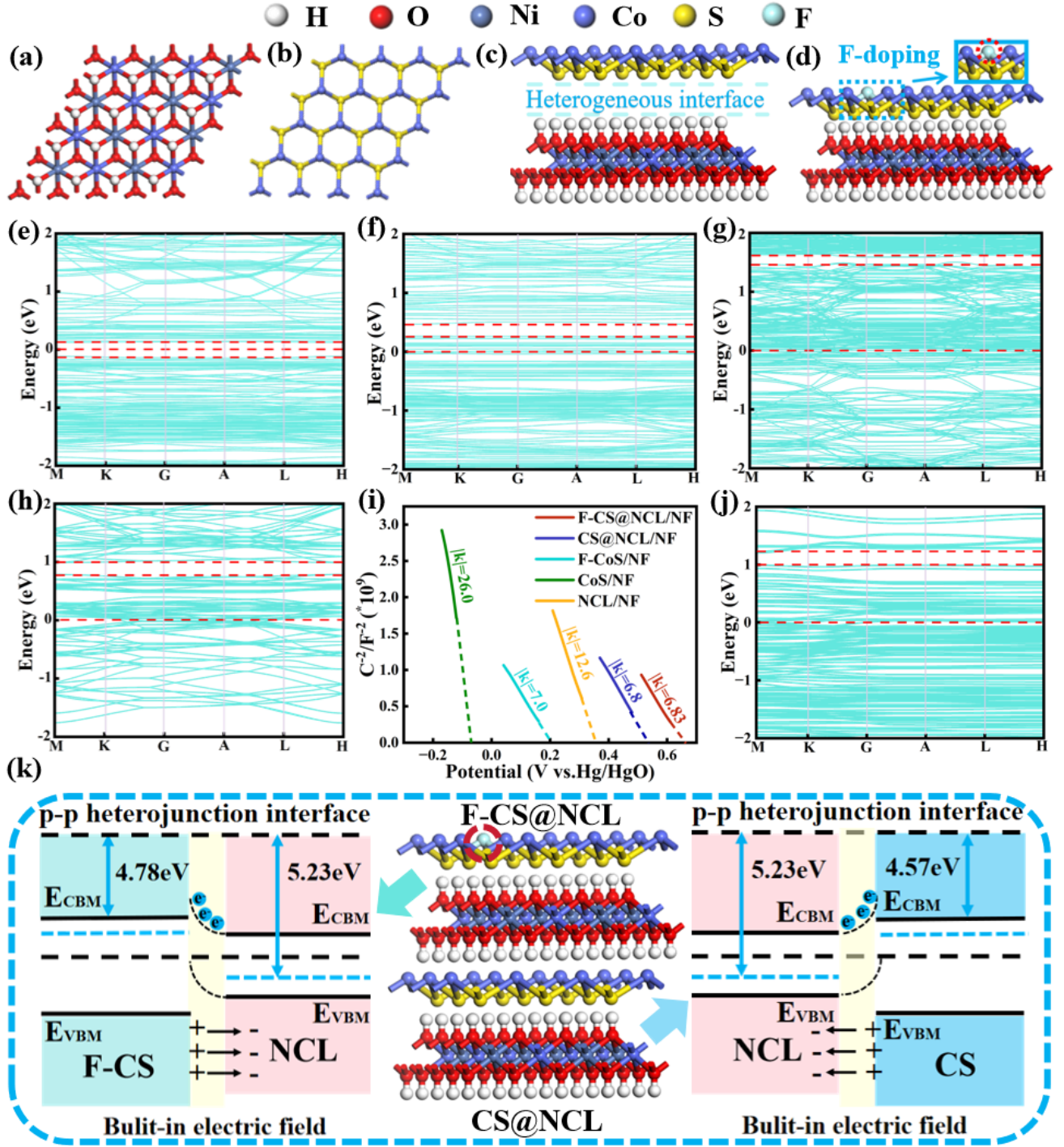


Fig. 2. Structural models of (a) NCL, (b) CoS, (c) CS@NCL, and (d) F-CS@NCL. Band structure plots of (e) NCL, (f) CS@NCL, (g) F-CS@NCL, (h) CoS, (j) F-CoS. (i) Absolute Value of Slope Comparison Diagram for Mott-Schottky curves of NCL/NF, CoS/NF, F-CoS/NF, CS@NCL/NF, F-CS@NCL/NF. (k) Illustration depicting the electron transfer mechanism. (Repeat each test three times to minimize errors).

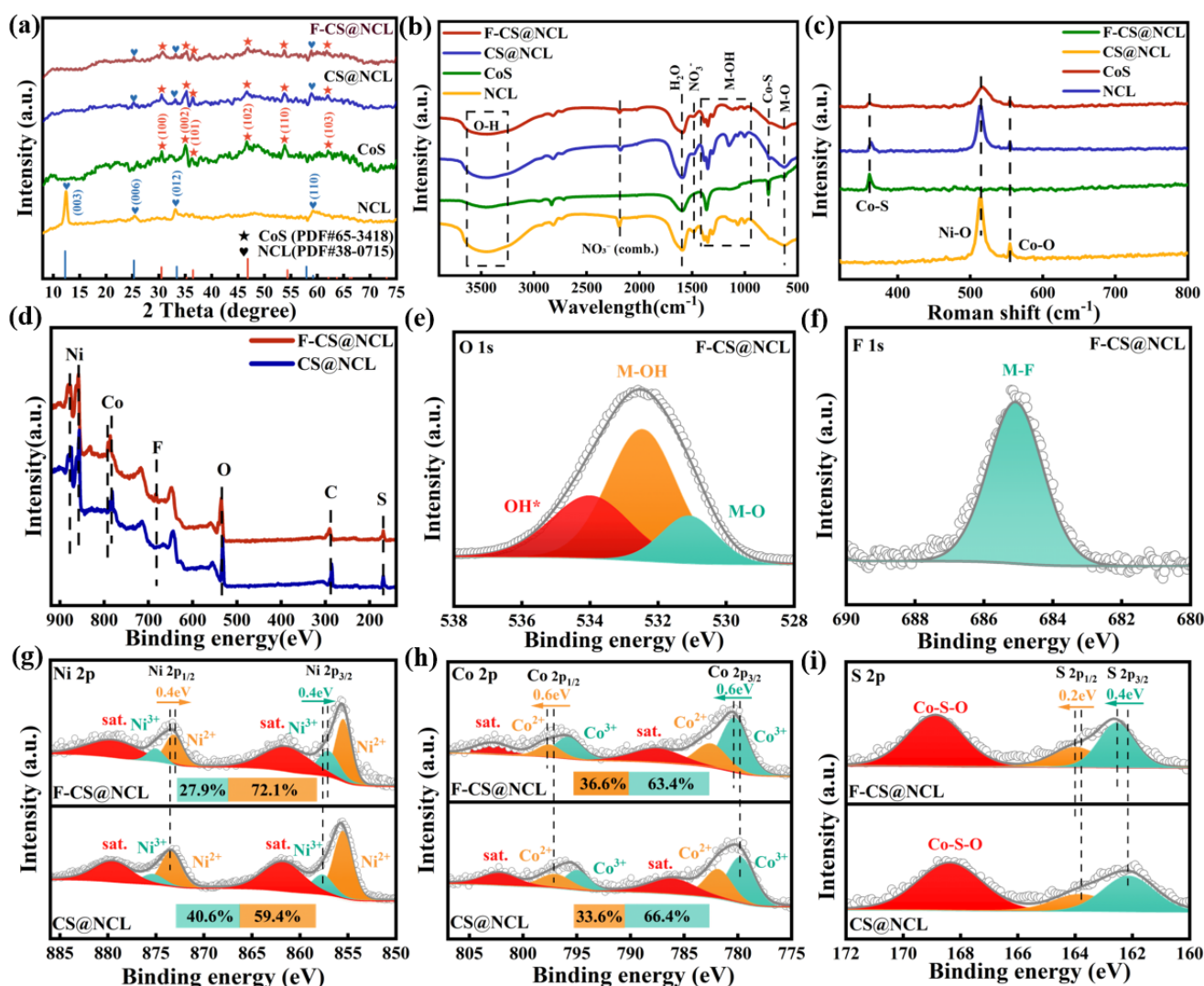


Fig. 3. (a) XRD profiles, (b) FTIR curves, and (c) Raman characterization of NCL/NF, CoS/NF, CS@NCL/NF, and F-CS@NCL/NF. (d) XPS of CS@NCL/NF and F-CS@NCL/NF. High-resolution XPS core-level spectra for (e) O 1s, (f) F 1s, (g) Ni 2p, (h) Co 2p, and (i) S 2p obtained from CS@NCL/NF and F-CS@NCL/NF. (Repeat each test three times to minimize errors).

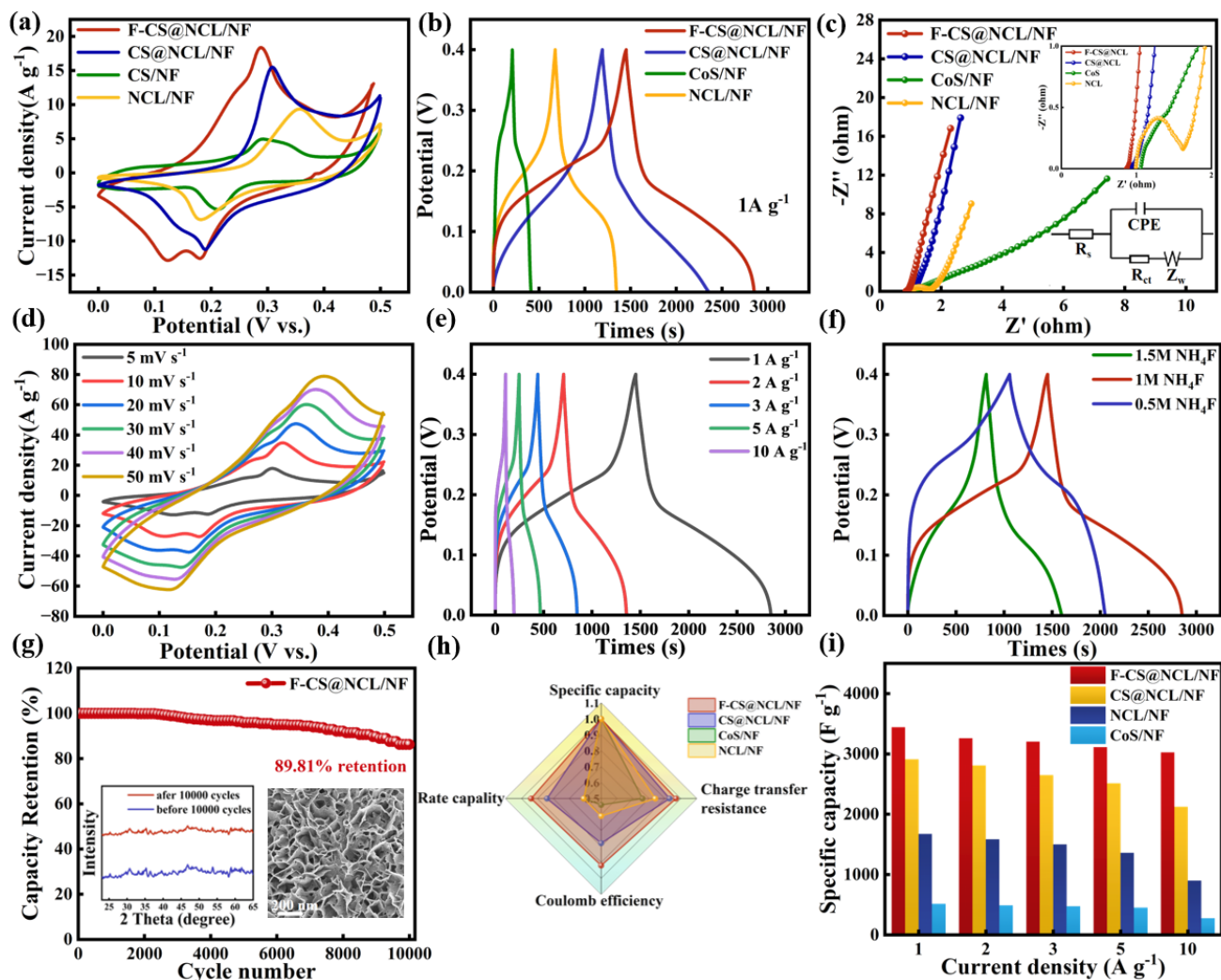


Fig. 4. (a) CV profiles comparing various electrodes at 5 mV s^{-1} . (b) GCD curves of the respective electrodes at 1 A g^{-1} . (c) Nyquist plots of the four electrodes. (d) CV curves for F-CS@NCL/NF across scan rates from 5 to 50 mV s^{-1} . (e) GCD curves of the F-CS@NCL/NF across different current densities. (f) Cycling stability of the F-CS@NCL/NF at 10 A g^{-1} . (g) Effect of Different Concentrations of Ammonium Fluoride on GCD Performance During F-Doping. (h) Specific capacity of the four electrodes as a function of current densities. (i) Specific capacity of the four electrodes as a function of current densities. (Repeat each test three times to minimize errors).

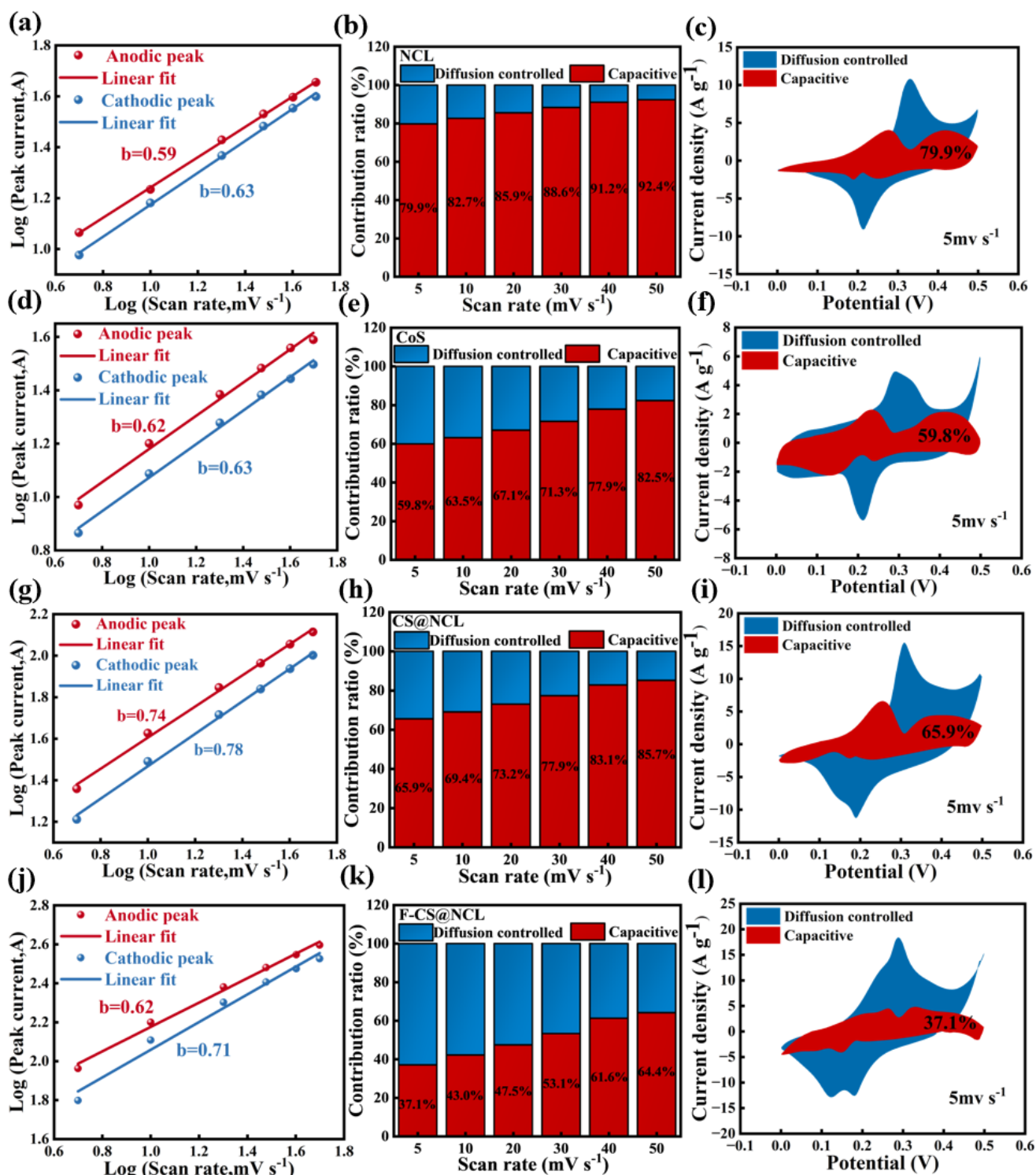


Fig. 5. (a, d, g, j) Linear fitting of $\log(i)$ versus $\log(v)$ for NCL/NF, CoS/NF, CS@NCL/NF, and F-CS@NCL/NF. (b, e, h, k) The respective contribution of the diffusion controlled and capacitive processes in total charge storage at different scan rates. (c, f, i, l) The corresponding CV curves showing the capacitive contribution and the diffuse controlled contribution at 5 mV s^{-1} . (Repeat each test three times to minimize errors).

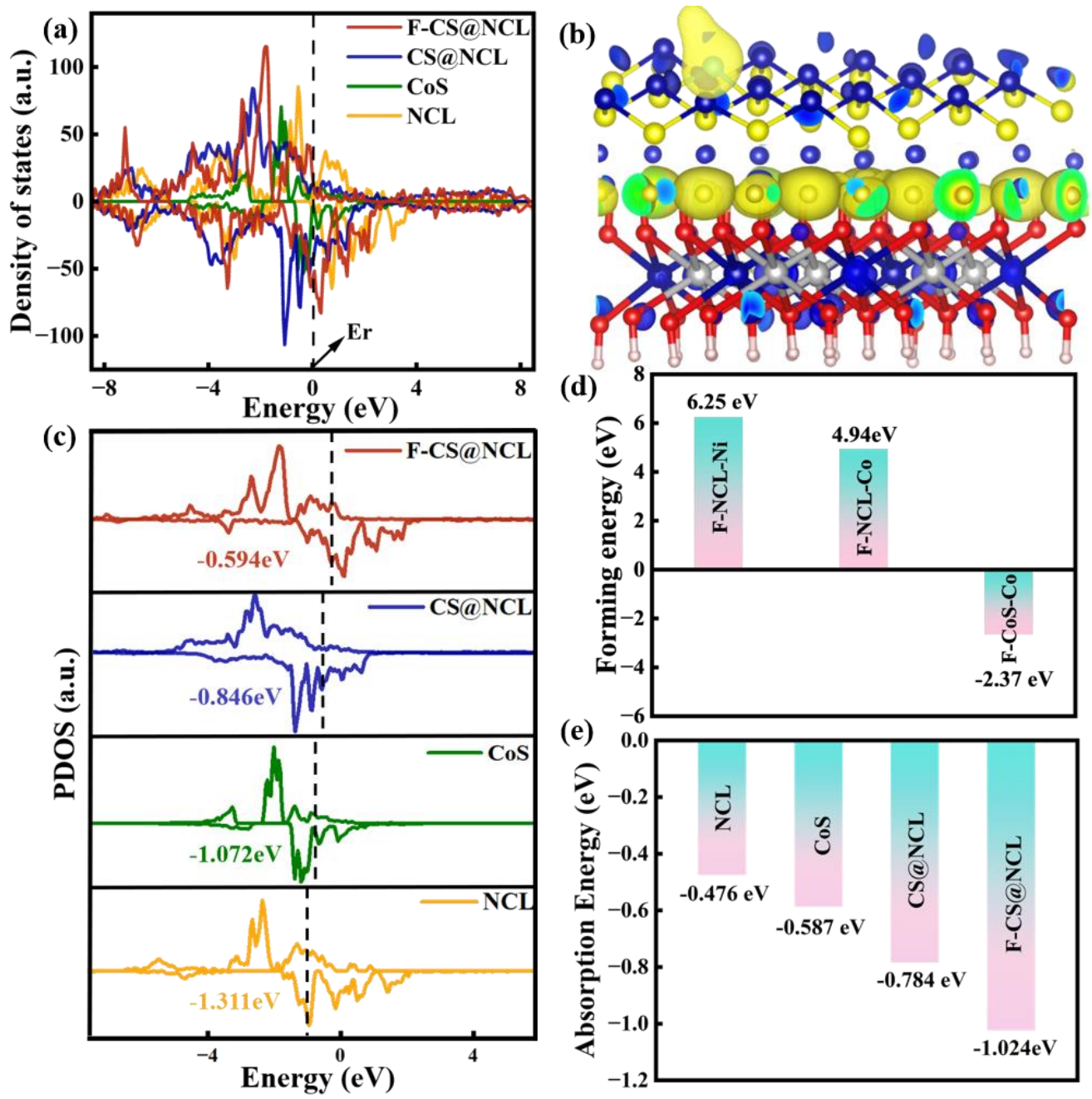


Fig. 6. (a) DOS of the NCL, CoS, CS@NCL and F-CS@NCL. (b) The differential density charge map of CS@NCL and F-CS@NCL. (c) PDOS of the NCL, CoS, CS@NCL and F-CS@NCL. (d) Formation energy of F-NCL-Co, F-NCL-Ni and F-CoS-Co. (e) H₂O adsorption energy of the NCL, CoS, CS@NCL and F-CS@NCL. (Repeat each test three times to minimize errors).

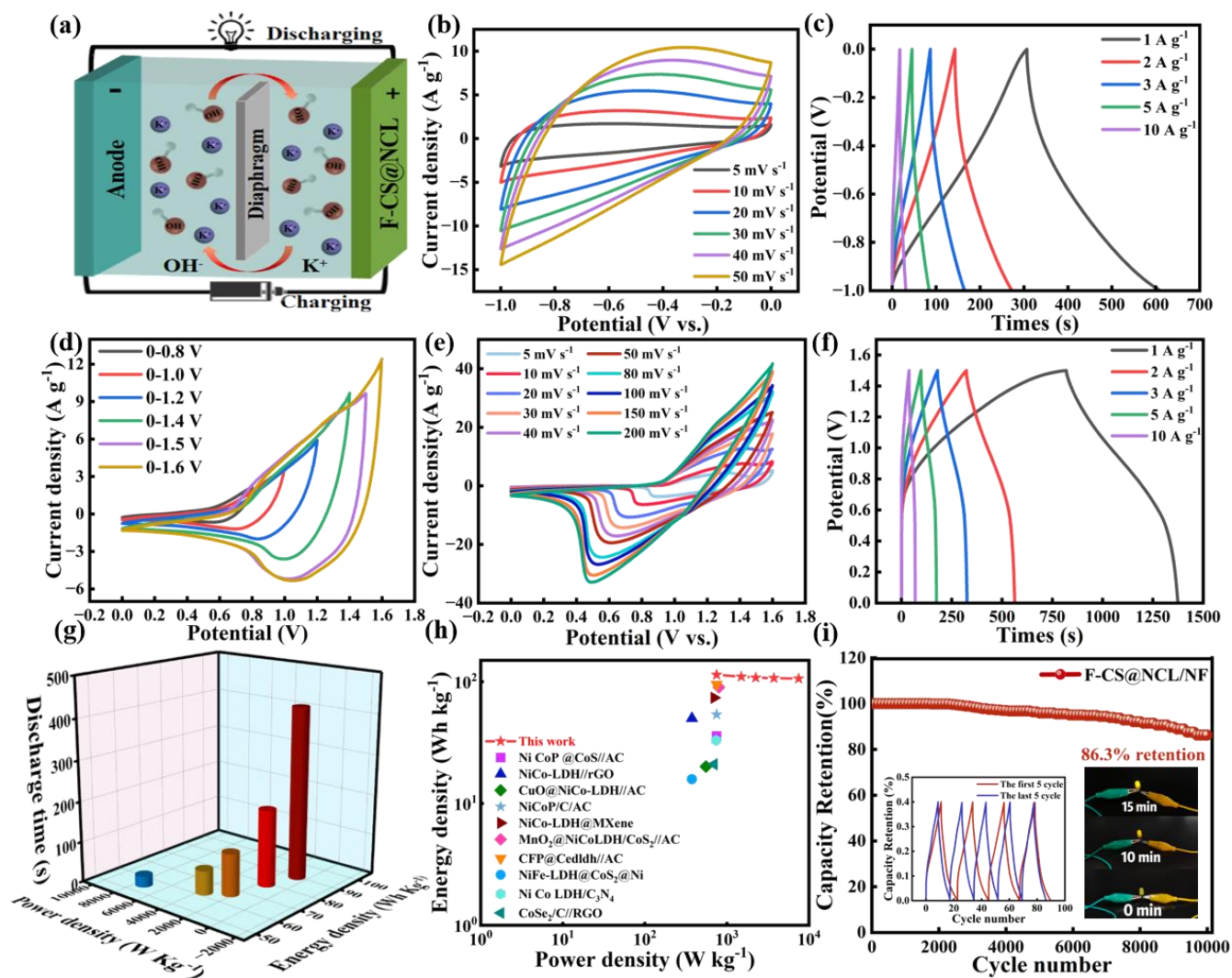


Fig. 7. (a) Energy storage mechanism of F-CS@NCL//AC HSC. (b) CV curves for the AC at multiple scan rates. (c) GCD responses of the AC under varied current densities. (d) CV curves for the HSC device across diverse voltage windows. (e) CV curves for the HSC within the potential range of 0–1.5 V. (f) GCD curves of the HSC at different current densities. (g) The relationship diagram of the power density, energy density and discharge time of the HSC equipment. (h) Ragone plots of the F-CS@NCL//AC HSC device and other SCs recorded in references. (i) Cycling stability and coulombic efficiency of the HSC equipment at 10 $A g^{-1}$. (Repeat each test three times to minimize errors).



Click here to access/download
Supporting material
Supporting-ZZH.docx

